

MAKTAB-E-DIGITAL SYSTEMS · LAHORE

MEDS-S1

An open RISC-V SoC platform for accelerator research

Platform Specification · Version 0.2 · DRAFT
Architecture, attachment interfaces, verification, and research methodology

Apache License 2.0 · Pre-RTL. Nothing herein is frozen until the Phase 0 design review.
Companion documents: INTERFACES (normative) · SCOPE CONTRACT · EXECUTION PLAN · GITHUB WORKFLOW

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Revision history

MEDS-S1 — Platform Specification

An open RISC-V SoC platform for accelerator research Maktab-e-Digital Systems (MEDS), UET Lahore · Apache-2.0

Version	0.2 — DRAFT for stakeholder review
Date	2026-08-03
Architect	Umer Shahid
Status	Pre-RTL. Freeze target: end of Phase 0
Companions	<code>INTERFACES.md</code> (normative), <code>SCOPE_CONTRACT.md</code> (v1.0 boundary), <code>../ADDENDUM.md</code> (rationale), <code>../EXECUTION_PLAN.md</code> , <code>../GITHUB_WORKFLOW.md</code>

Part I — The platform

1. Purpose

MEDS-S1 is a RISC-V system-on-chip platform whose purpose is **to be attached to**.

The scalar core is deliberately conventional. The value is in five things that are hard to build and harder to retrofit:

1. **Frozen, versioned attachment interfaces** — a tightly-coupled instruction-extension port (MXIF) and a loosely-coupled memory-mapped accelerator socket, both specified in `INTERFACES.md`.
2. **A generator** that turns one `soc.yaml` into RTL, linker script, device tree, C headers, OpenOCD config and documentation, so adding a peripheral or an accelerator is a config change.
3. **A verification harness** in CI that tells a second-year student in twenty minutes whether their change broke the ISA.
4. **A BSP and debug path** that make running and debugging arbitrary C on real hardware a ten-second operation with no resynthesis.
5. **Measurement infrastructure** that makes every attached accelerator's performance claim comparable to every other one's.

1.1 The research context

MEDS-S1 exists to support MS and PhD work in **Edge AI** and **healthcare electronics**. Those domains share a shape: modest models, hard latency and energy constraints, and a need to prove that a proposed accelerator actually helps *in a system*, not in isolation on a spreadsheet.

The recurring failure mode in accelerator research is a synthesised block with an impressive TOPS/W number that was never attached to a processor, never fed by a real memory system, and never ran a real workload end to end. Reviewers know this and discount accordingly.

MEDS-S1's research proposition is: **give every thesis student a working SoC on day one, so their contribution is the accelerator and the measurement, not the plumbing**. A student should be able to attach a design, run an ECG classifier on it, and report end-to-end latency, memory traffic, and speedup against a scalar baseline — within a semester.

1.2 What MEDS-S1 is not

- Not a high-performance core. It is in-order, single-issue, and will lose every IPC comparison to CVA6 or Rocket. That is a deliberate allocation of the innovation budget (`SCOPE_CONTRACT.md` §3).
 - Not a product for silicon. It is written ASIC-clean so the option stays open, but v1.0 targets FPGA.
 - Not a replacement for CVA6 or Rocket if what you need is a mature Linux-capable core today.
-

2. Stakeholders and use cases

#	Stakeholder	What they need	Primary interface
S1	MS student attaching a DL accelerator	a working SoC, a socket, a driver template, a baseline to beat	Accelerator socket (§20), <code>libs1_perf</code>
S2	PhD student doing microarchitecture research	modifiable core RTL, a golden reference, formal + coverage, comparable numbers	S1-Core RTL, RVFI (§28)
S3	PhD student doing custom-ISA research	opcode space, an extension checklist, toolchain path, arch-tests	MXIF (§19), extension registry
S4	Undergraduate contributor	a bounded first task, docs, fast feedback	<code>soc.yaml</code> , HAL, testbenches
S5	Healthcare/Edge-AI applications researcher	run my model, measure it, don't make me learn Verilog	BSP, semihosting, workload suite (§32)
S6	External adopter / industry evaluator	compliance evidence, a licence, documentation	Release artefacts (§28.6)
S7	Faculty supervisor	comparable results across students, continuity across cohorts	Comparison methodology (§34)

2.1 Driving use case, stated end to end

A second-year MS student proposes a sparsity-aware INT8 convolution engine for ECG arrhythmia classification on an edge device.

Week 1 — clones `meds-s1`, runs `make run BOARD=verilator PROG=examples/ecg_cnn.elf`, gets a scalar baseline: 41 ms/inference, 12.3 M cycles, 2.1 MB of DRAM traffic. **Weeks 2-10** — develops the engine against the socket testbench, in Verilator, without an FPGA. **Week 11** — sets `accel0: { ip: sparse_conv }` in `soc.yaml`, rebuilds, runs on KC705. **Week 12** — reports 3.8 ms/inference, 0.9 M cycles, 1.4 MB DRAM traffic, with the accelerator busy 78% of the time and DMA stalled 14% — and can say *why*, because the counters distinguish them.

Nothing in weeks 1, 11 or 12 required modifying core RTL, writing a bus adapter, or porting a toolchain. That is the entire point of the platform.

3. Requirements

Numbered, testable, and traceable to a verification item. `SCOPE_CONTRACT.md` §6 is the acceptance gate; this is the requirement list behind it.

3.1 Functional

ID	Requirement	Verified by
FR-1	Implements RV64IMAC_Zicsr_Zifencei_Zicbom_Zicboz, M and U mode	RISCOF vs Sail
FR-2	S-mode CSRs, delegation and privilege FSM present; behaviour stubbed in v1.0	directed tests
FR-3	16 PMP regions with standard granularity and locking	directed + arch-tests
FR-4	PMA checks enforced per region attribute set (§11)	directed tests
FR-5	Precise exceptions at instruction granularity, incl. offloaded instructions	co-simulation
FR-6	MXIF-1.0 port per INTERFACES.md §1	socket conformance TB
FR-7	≥1 accelerator socket per INTERFACES.md §4, count set by soc.yaml	socket conformance TB
FR-8	RISC-V Debug Module: halt, resume, step, 2 triggers, memory and register access	OpenOCD test suite
FR-9	Load and run an arbitrary ELF over JTAG with no resynthesis	acceptance test
FR-10	Semihosting: open / read / write / close / seek to the host filesystem	acceptance test
FR-11	CLINT + PLIC; all interrupt sources level-sensitive	directed tests
FR-12	soc.yaml generates RTL, link.ld, C headers, DTS, OpenOCD cfg, docs	generator golden tests
FR-13	RVFI + RVFI-V trace port at retire	co-simulation
FR-14	mcycle, minstret, mhpcounter3–15 with programmable events	directed tests
FR-15	Boots from ROM; second-stage load from UART/SD/QSPI	acceptance test

3.2 Non-functional

ID	Requirement	Verified by
NFR-1	≥ 50 MHz on Kintex-7 (KC705) at the S1-Base config	synthesis + timing report
NFR-2	≥ 1.0 CoreMark/MHz	benchmark CI
NFR-3	Full Verilator regression < 20 min on the CI runner	CI timing
NFR-4	Verible lint clean, zero unjustified waivers	CI
NFR-5	No inferred latches; no FPGA primitives in core RTL; all memory via <code>meds_sram_wrapper</code>	lint + review
NFR-6	Single clock domain in the core; CDC only in named synchroniser modules	CDC lint
NFR-7	Every module has a <code>README.md</code> stating its interface contract	CI doc check
NFR-8	A new board port touches exactly 4 files	review
NFR-9	An S4-tier contributor can add a peripheral and see it from C in < 1 day, docs only	onboarding test each cohort
NFR-10	Benchmark regression > 3% blocks merge without written justification	CI

3.3 Interface requirements

Normative interface requirements live in `INTERFACES.md` (R1.1–R1.11, P1–P5) and are incorporated here by reference. They are the requirements that must not change; everything above may.

4. System overview

4.1 Top-level block diagram — S1-AI configuration

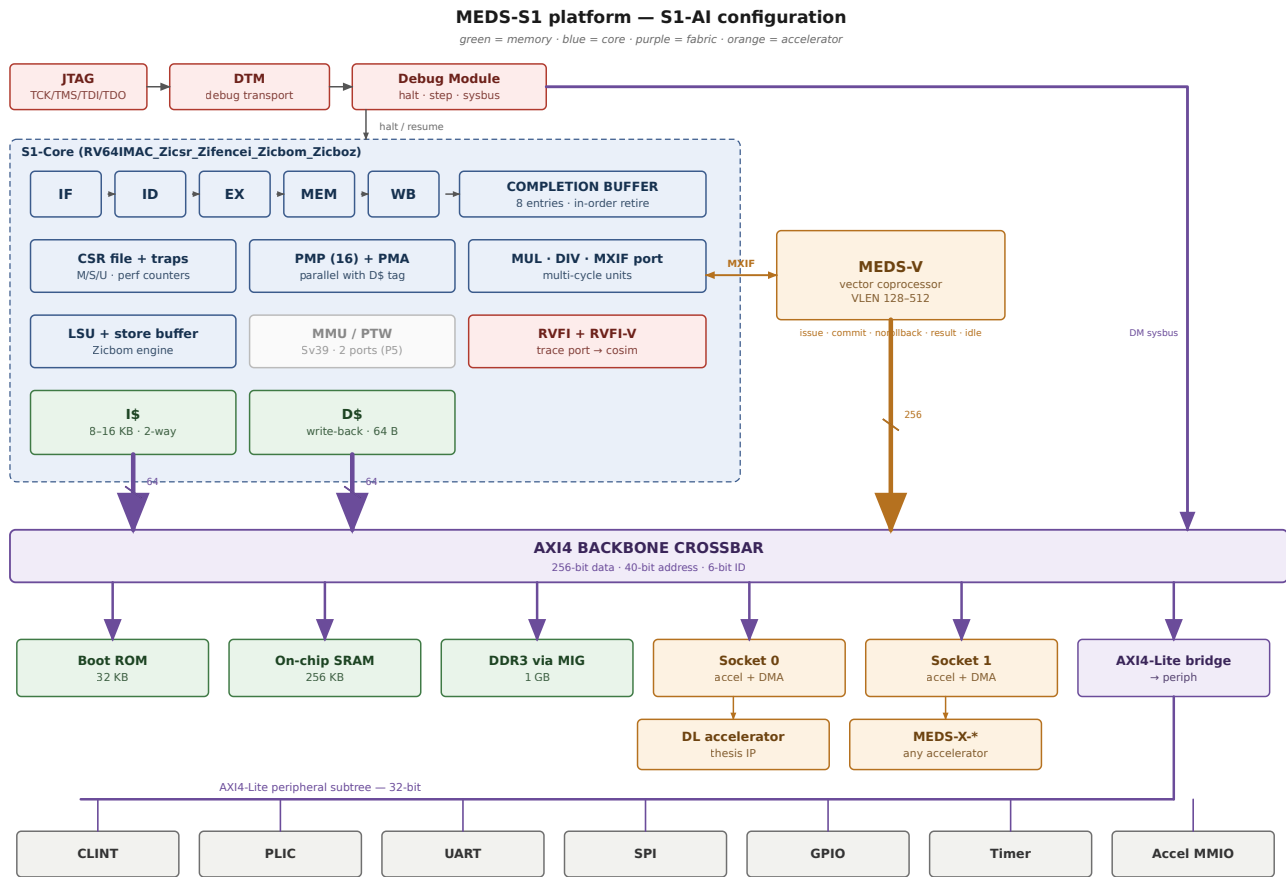
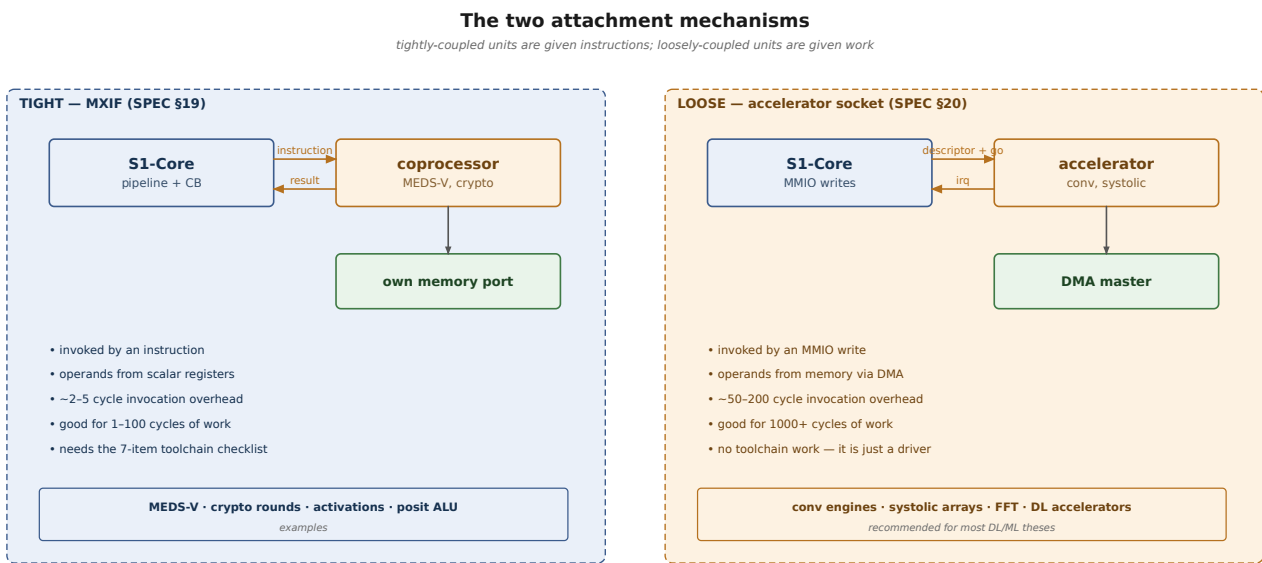


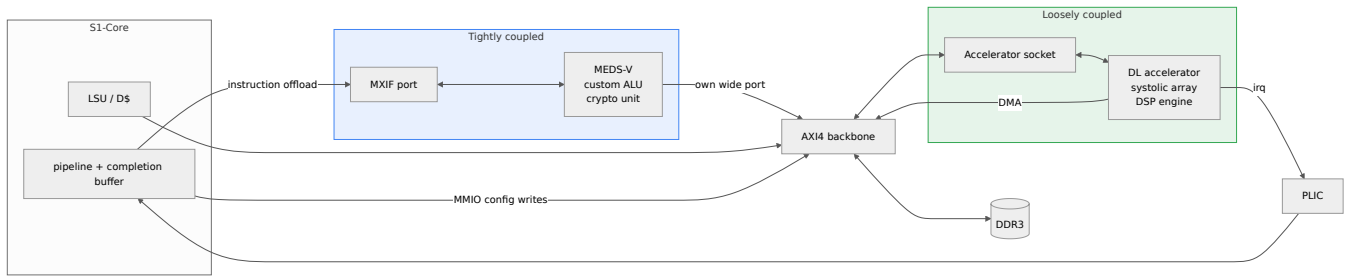
Figure 1 — MEDS-S1 top-level block diagram, S1-AI configuration.

4.2 Where the two attachment mechanisms sit



The common mistake: attaching a convolution engine over MXIF because instructions feel elegant.
A convolution runs for thousands of cycles and streams megabytes. It does not want to be an instruction.

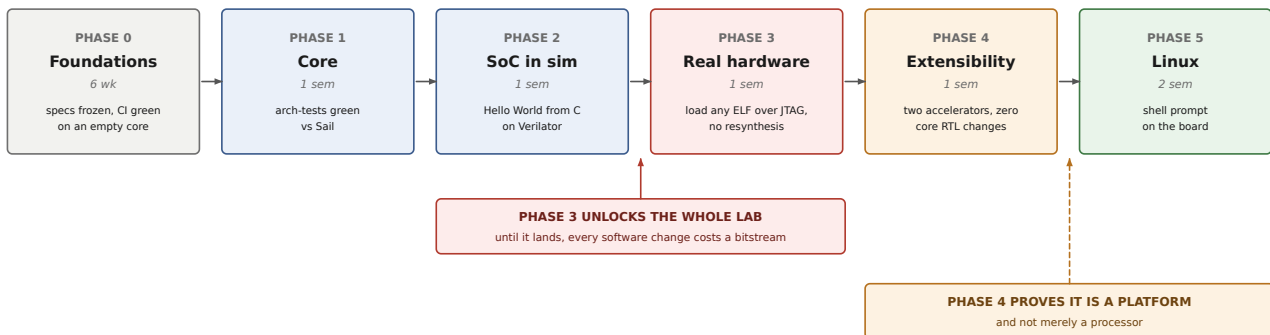
Figure 2 — The two attachment mechanisms compared.



The distinction in one sentence: **tightly-coupled units are given instructions; loosely-coupled units are given work.**

5. Naming, versions and configurations

Phase plan — each phase ends with something that runs



Critical path: WP0 → WP3 → WP5 (completion buffer + MXIF) → WP15 (debug). WP5 is both on the critical path and the hardest package.

Figure 3 — Phase plan: each phase ends with something that runs.

5.1 Names

Name	Meaning
MEDS-S1	this platform: core, fabric, generator, BSP, CI, board ports
S1-Core	the scalar RV64 CPU inside MEDS-S1
MEDS-V	the RVV vector coprocessor, attached over MXIF
MEDS-X-<name>	any accelerator built for MEDS-S1
MXIF	MEDS eXtension InterFace, a profile of OpenHW CV-X-IF
MEDS-S2	reserved for the successor platform

5.2 Versioning

Semantic versioning on the platform (`vMAJOR.MINOR.PATCH`) and independently on each interface (`MXIF-1.0`, `SOCKET-1.0`). A platform release pins every submodule hash in `platform.lock`.

`MEDS_S1_PLATFORM_VERSION` is readable from the boot ROM and from a read-only CSR, so software can identify its hardware at runtime.

5.3 Configurations

Four named build configurations. Every one is elaborated in CI on every merge; **a config that is not in CI does not exist.**

	S1-Nano	S1-Base	S1-AI	S1-Linux
Purpose	teaching, CI smoke, tiny FPGA	the default	accelerator research	Phase 5
ISA	RV64IMC	RV64IMAC_Zicsr_Zifencei_Zicbom	+ Zicboz	+ S-mode, Sv39
Privilege	M	M, U	M, U	M, S, U
I\$ / D\$	none (TCM)	8 KB / 8 KB 2-way	16 KB / 16 KB 2-way	16 KB / 16 KB 2-way
MMU	no	no	no	Sv39, 32-entry TLB
PMP regions	0	16	16	16
Backbone	64-bit	64-bit	256-bit	256-bit
MXIF	no	yes	yes + MEDS-V	yes + MEDS-V
Sockets	0	1	2	2
Debug	no	yes	yes	yes
Memory	64 KB SRAM	SRAM + DDR	SRAM + DDR	SRAM + DDR
Target board	Verilator	Verilator, KC705	KC705	KC705
Est. LUTs	~8 k	~25 k	~45 k (excl. accel)	~55 k

(LUT figures are design targets for Phase-0 planning, not measurements. They are replaced with synthesis results at the end of Phase 2 and the estimates are kept alongside for calibration.)

Part II — The S1 core

6. Pipeline organisation

Five stages, single issue, in order. Multi-cycle work leaves the main pipe at EX and returns through a completion buffer, which is what makes accelerator attachment possible without stalling everything.

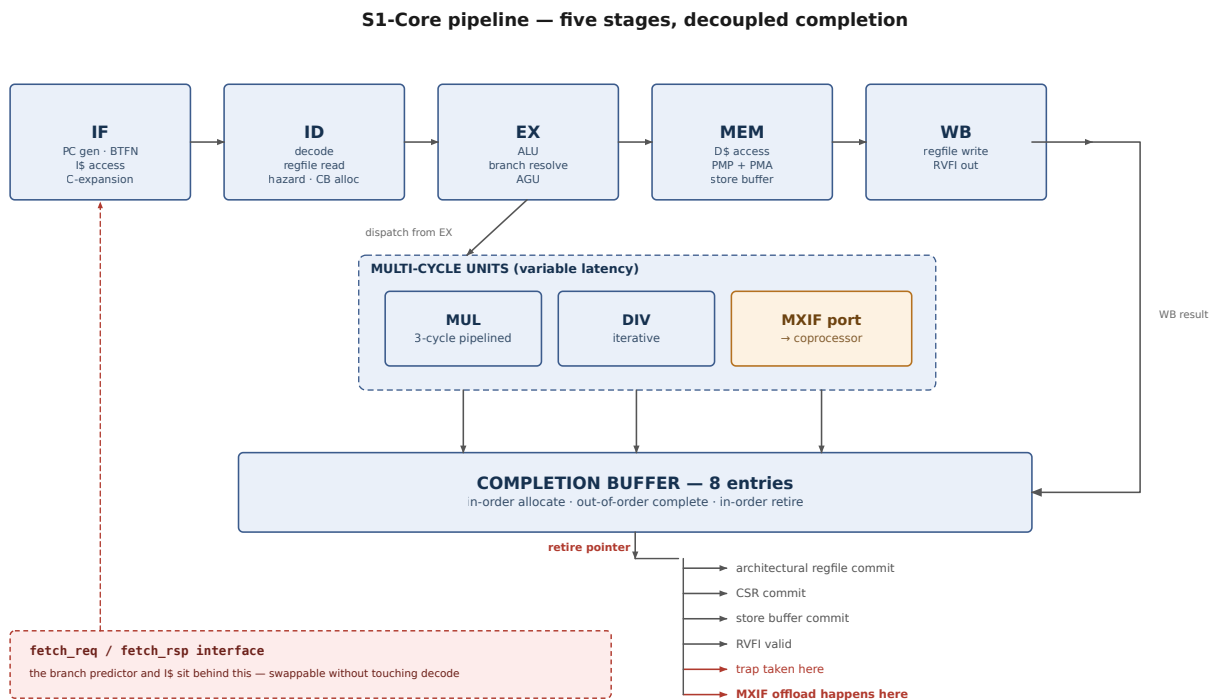


Figure 4 — S1-Core pipeline organisation and the completion buffer.

The two structural decisions that everything else follows from:

1. **The frontend sits behind a `fetch_req / fetch_rsp` interface.** The branch predictor and the I\$ can be replaced without touching decode. This is what turns “add a gshare predictor” into a bounded, measurable v2 thesis project with a v1 baseline, rather than a pipeline rewrite.
2. **All architectural state updates happen at the retire pointer**, not at WB. This gives precise exceptions with variable-latency units, and it is the only reason an accelerator can complete out of order without stalling the world.

7. Stage by stage

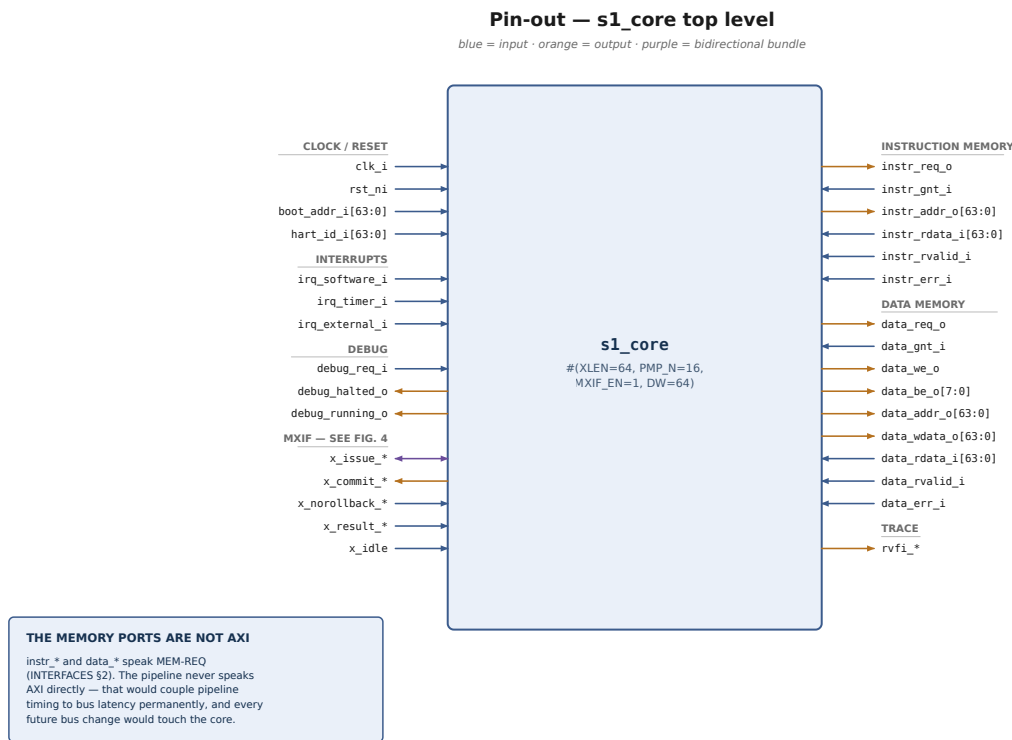


Figure 5 — s1_core top-level pin-out.

7.1 IF — instruction fetch

- PC generation; next-PC from sequential, BTFN prediction, branch resolution, or trap vector.
- Static **BTFN** (backward-taken, forward-not-taken) prediction — a sign-bit test on the branch immediate, no storage.
- I\$ access via `fetch_req/fetch_rsp`; 32-bit aligned fetch, with a 16-bit realign buffer for `C`.
- Misaligned fetch across a cache line is handled by the realign buffer, not by the pipeline.

C extension note. Compressed instructions are expanded to 32-bit form at the IF/ID boundary. Everything downstream — including MXIF — sees only 32-bit encodings. This is why MXIF-1.0 omits CV-X-IF's compressed channel (`SCOPE_CONTRACT.md` §3).

7.2 ID — decode

- Instruction decode to a single `decoded_op_t` control bundle. **One decoder, one struct** — no control signals threaded individually through the pipeline.
- Register file read (2 ports), immediate generation, hazard detection, forwarding-source selection.
- Allocates a completion buffer entry. **Stalls if the CB is full.**
- Classifies the instruction as: main-pipe, multi-cycle (MUL/DIV), MXIF-candidate, CSR, or system.
- An MXIF-candidate is any instruction the base decoder does not recognise, plus opcodes explicitly routed to a coprocessor by `soc.yaml` (e.g. `0x57`, `0x07`, `0x27` when MEDS-V is present).

Design note. The decoder does *not* reject unknown instructions. It marks them MXIF-candidate and lets the coprocessor accept or reject at the retire point. Illegal-instruction is raised only if no coprocessor accepts, or if none is present. This single rule is what makes custom instructions possible without touching the decoder.

7.3 EX — execute

- ALU: add/sub, logic, shift, compare, `lui/auipc`.
- Branch resolution and target computation; a mispredict flushes IF and ID.
- Address generation for loads and stores.
- Dispatch to MUL/DIV.
- CSR read-modify-write is computed here, **committed at retire**.

7.4 MEM — memory

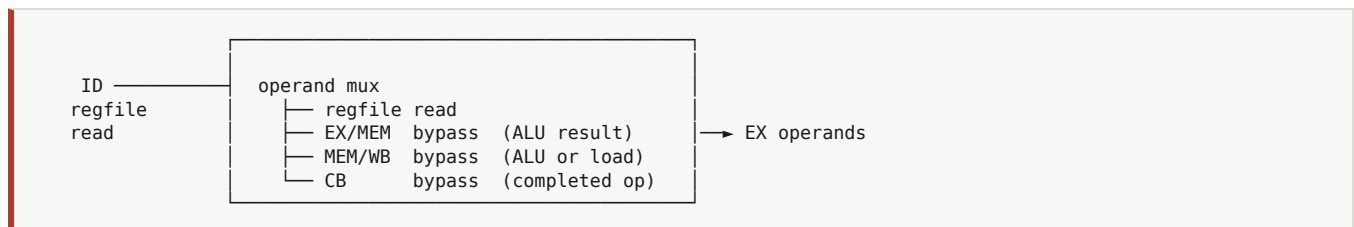
- D\$ access; PMP and PMA checks in parallel with tag lookup (§11).
- Store buffer entry allocation. Stores commit to the buffer at retire, never before.
- `Zicbom` cache-block operations issue here.
- AMO and LR/SC sequencing.

7.5 WB / retire

- Register file writeback for main-pipe instructions.
- Completion buffer update.
- **Retire pointer advance** — the architectural commit point (§9).
- RVFI trace emission (§28.2).

8. Hazards, forwarding and stalls

8.1 Forwarding network



Four sources. The fourth — forwarding from a completed-but-not-retired completion buffer entry — is what stops a multi-cycle result from stalling its consumer until retire.

8.2 Hazard table

Hazard	Detected in	Resolution	Penalty
RAW, producer in EX	ID	forward EX/MEM	0
RAW, producer in MEM	ID	forward MEM/WB	0
RAW, load in MEM (load-use)	ID	stall 1	1
RAW, producer is MUL/DIV in flight	ID	stall until CB entry completes, then forward	variable
RAW, producer is an MXIF op with writeback	ID	stall until <code>x_result_valid</code>	variable
Branch mispredict	EX	flush IF, ID	2
Trap / interrupt	retire	flush all, redirect to vector	3-4
<code>fence.i</code>	retire	flush all, invalidate I\$	I\$ flush time
<code>fence</code>	retire	wait for store buffer drain and <code>x_idle</code>	variable
CB full	ID	stall	variable
Scalar memory op while coprocessor memory outstanding	MEM	stall (MXIF \$1.5)	variable
CSR write to a CSR read by an in-flight instruction	ID	serialise: stall until CB empty	variable

The last row is a deliberate simplification: **CSR writes serialise the pipeline**. CSR writes are rare and the alternative is a renaming structure nobody needs. Document it, measure it, move on.

9. Completion buffer and retire

The completion buffer (CB) is a small circular FIFO. Every instruction allocates an entry at ID, in order; entries complete out of order; the head retires in order.

9.1 Entry format

```
typedef struct packed {
    logic        valid;
    logic        done;           // result available
    logic        norollback;     // cannot fault any more (MXIF R1.9; always 1 for main-pipe)
    logic [63:0] pc;
    logic [31:0] instr;
    logic [4:0]   rd;
    logic        rd_we;
    logic [63:0] result;
    logic        exc;
    logic [5:0]   exccode;
    logic [63:0] exctval;
    logic        is_mxif;
    logic [3:0]   mxif_id;
    csr_update_t csr;
    store_buf_ptr_t stp;         // store buffer entry, if any
} cb_entry_t;
```

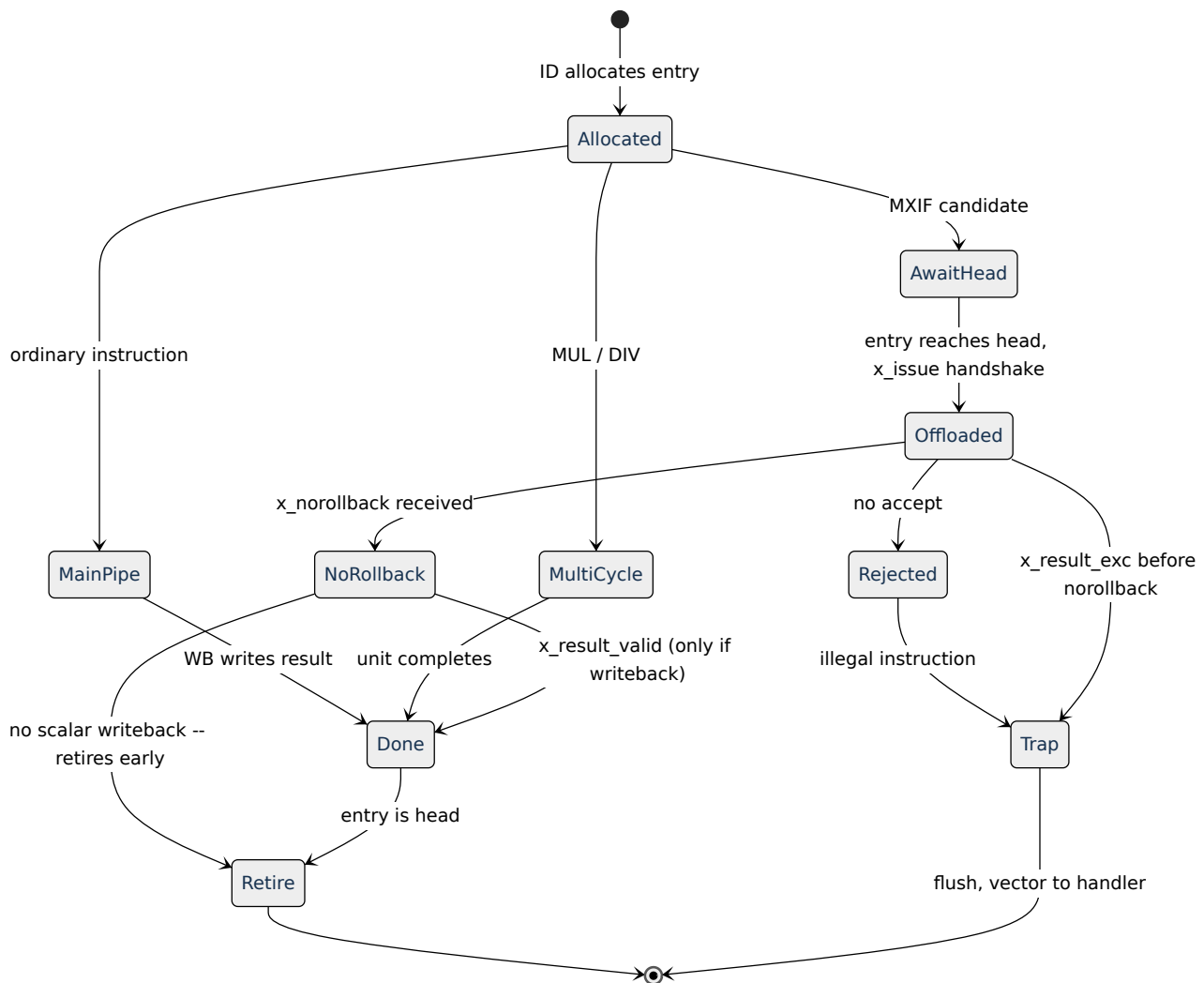
9.2 Retire rules

An entry at the head retires when:

```
valid && norollback && (done || (is_mxif && !rd_we))
```

Read that carefully — it is the heart of §19’s decoupling. A main-pipe instruction needs `done`. An MXIF instruction that writes no scalar register needs only `norollback`, so it retires while the coprocessor is still executing. An MXIF instruction that *does* write a scalar register needs both.

On retire: 1. Architectural register file write, if `rd_we`. 2. CSR commit, if any. 3. Store buffer entry marked committed (it may drain to memory later). 4. RVFI channel driven. 5. `minstret++`. 6. If `exc` — flush everything, take the trap, do not perform 1–3. 7. If the entry is an unstarted MXIF-candidate — **offload now** (R1.1).



9.3 Why eight entries

Enough to cover an iterative divide (~34 cycles at worst) plus in-flight main-pipe work without the CB becoming the bottleneck; small enough that the head-comparison logic is not on the critical path. It is a parameter. If a workload shows it binds, raise it and report the measurement — do not raise it on intuition.

9.4 Deadlock argument

The head always eventually retires:

- Main-pipe instructions complete in bounded time.
- MUL/DIV complete in bounded time.

- MXIF: the coprocessor must assert `x_issue_ready` or `x_issue_accept=0` in bounded time (`INTERFACES.md` conformance requirement), and must assert `x_norollback` in bounded time after accept. Both are conformance-testable properties, checked by the socket conformance testbench.

This paragraph is a verification requirement, not prose. Each claim maps to an SVA property in `verif/formal/cb_liveness.sv`.

10. Privilege, CSRs and traps

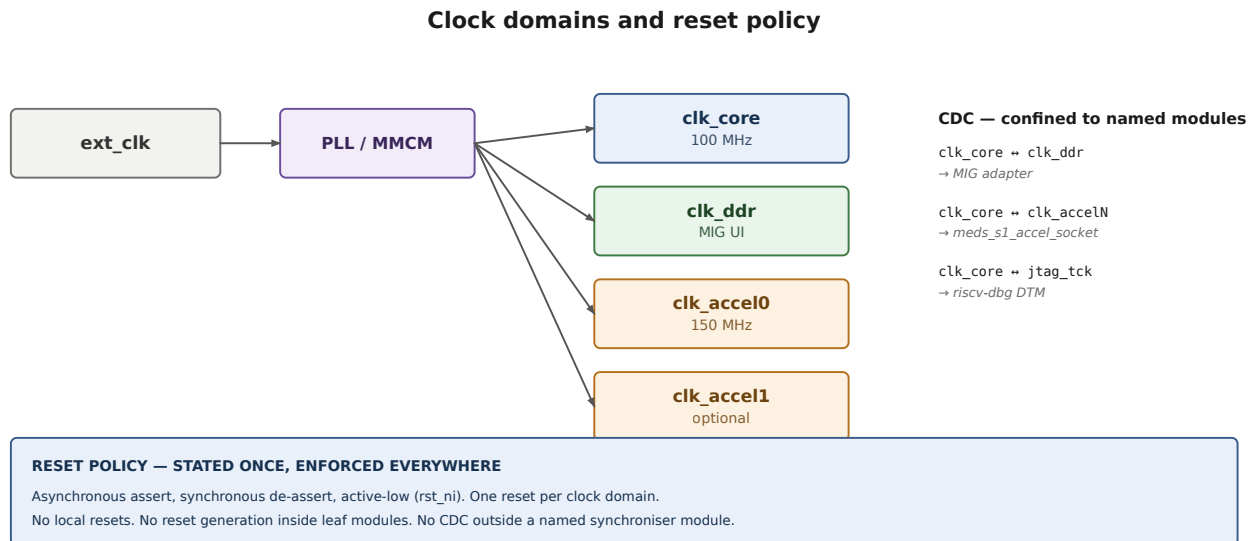
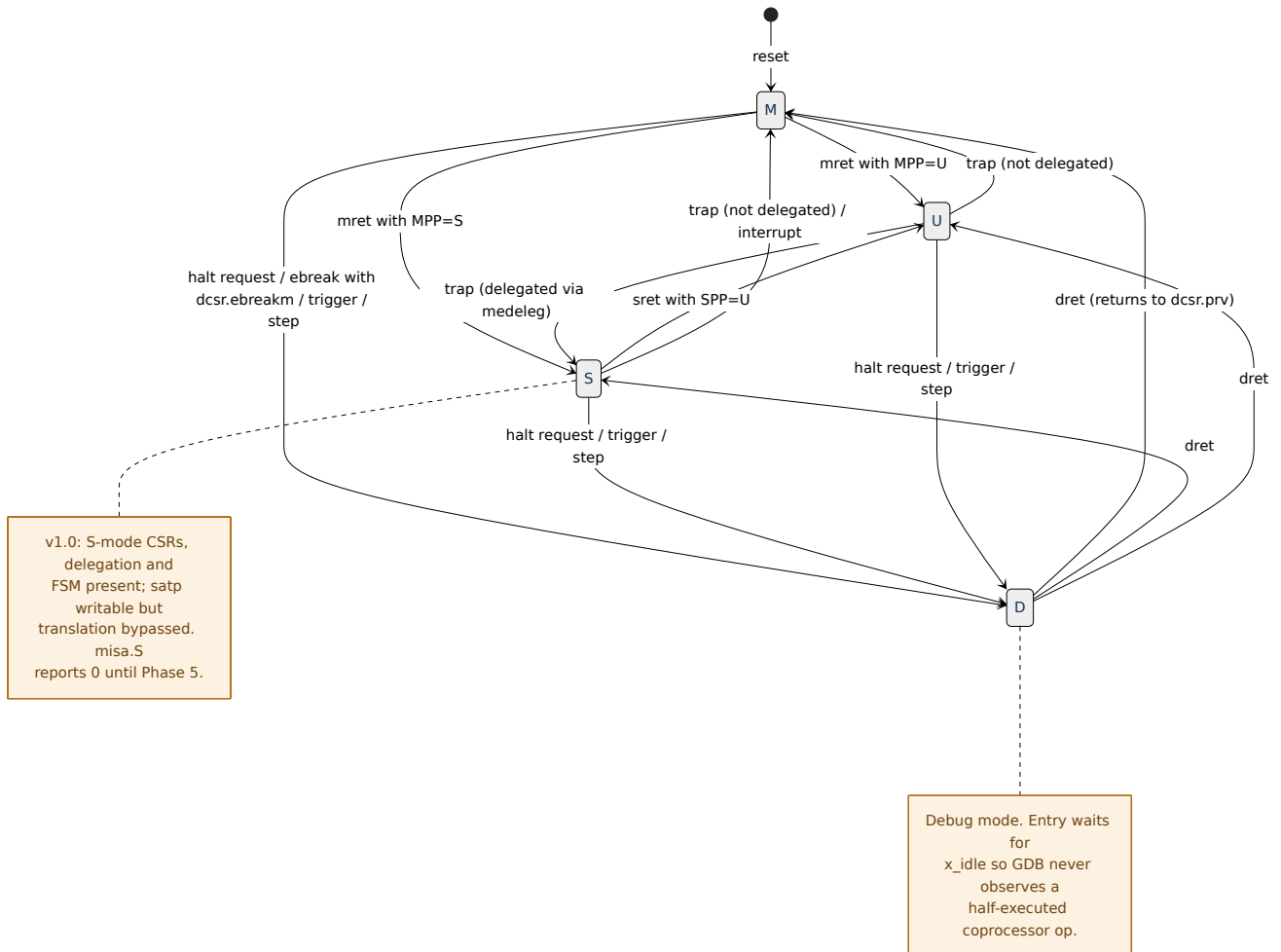


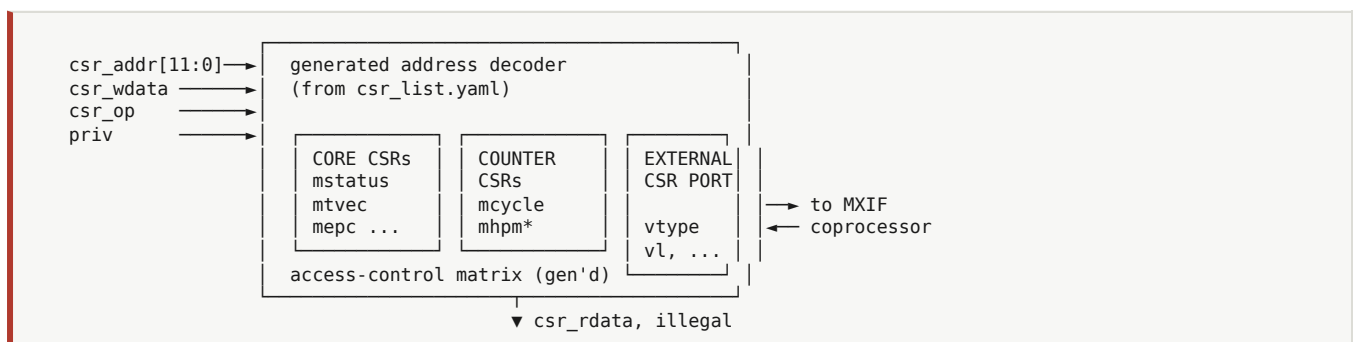
Figure 6 — *Clock domains and reset policy (§25), shown here because the privilege FSM and debug entry are clock-domain sensitive.*

10.1 Privilege state machine



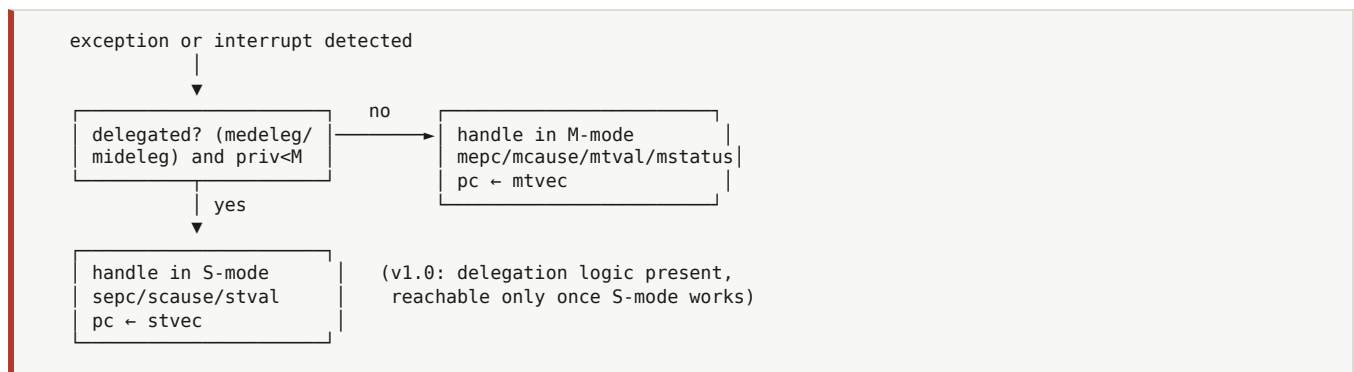
10.2 CSR file structure

Not a hand-written case statement. The CSR file is a generated, address-decoded structure with three classes of entry, because MEDS-V needs seven CSRs of its own and every future extension will need more:



The external CSR port is mandatory in v1.0. A coprocessor owns its own CSRs; the core routes reads and writes to it and applies the standard privilege check. Without this port, adding MEDS-V means editing the core's CSR file — which is exactly the coupling this platform exists to prevent.

10.3 Trap flow



Priority order — fixed and testable: debug halt > interrupts (external > software > timer, M > S) > synchronous exceptions in the standard RISC-V order.

11. PMP and PMA checking

Physical memory map and attributes					
generated from soc.yaml into the decoder, linker script, device tree and C headers					
base	region	size	cacheable	ordering	
0x1_0000_0000	DRAM uncached alias	1 GB	no	strong	→
0x8000_0000	DRAM (cached)	1 GB	yes	rvwmo	
0x4000_0000	On-chip SRAM	256 KB	yes	rvwmo	
0x2001_0000	Socket 1 MMIO	64 KB	no	strong	
0x2000_0000	Socket 0 MMIO	64 KB	no	strong	
0x1000_0000	Peripherals	256 MB	no	strong	
0x0C00_0000	PLIC	4 MB	no	strong	
0x0200_0000	CLINT	64 KB	no	strong	
0x0000_1000	Boot ROM	32 KB	I-only	rvwmo	
0x0000_0000	Debug ROM	4 KB	no	rvwmo	

PMA ATTRIBUTES CARRIED PER REGION

cacheable · idempotent · ordering · atomicity · alignment · widths

P1 a non-idempotent region is never speculated, prefetched or replayed

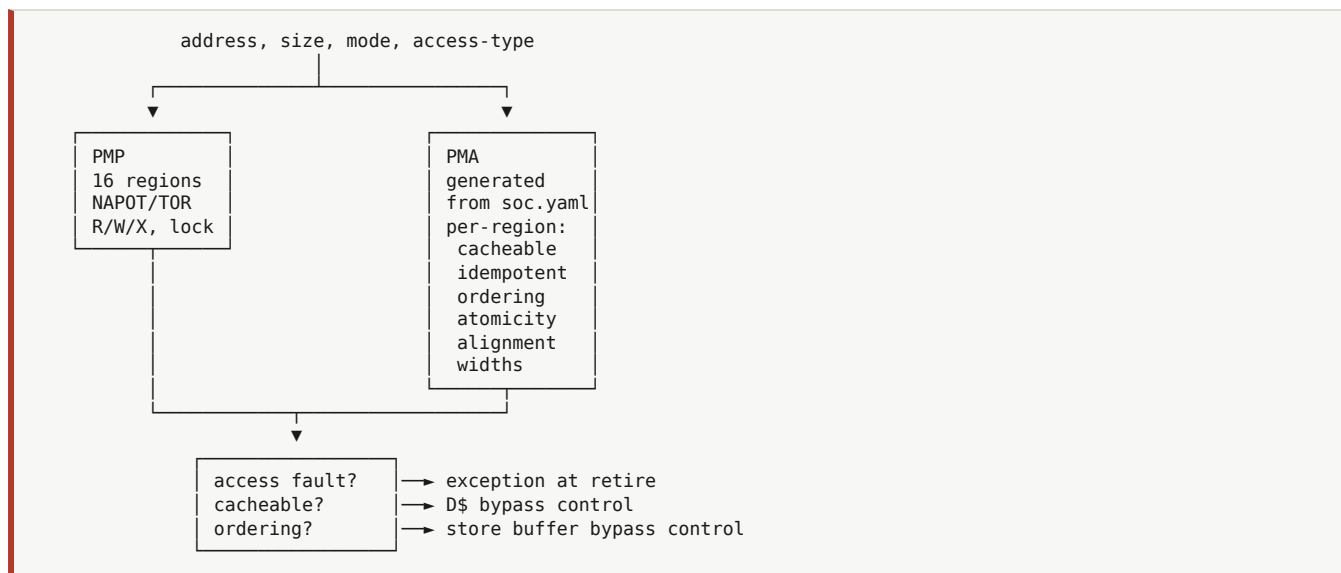
P2 order:strong regions are accessed in program order without a fence

P3 an alignment or width violation faults, never truncates silently

THE UNCACHED ALIAS

The same physical DRAM, mapped twice. Costs one address-decode bit and gives every accelerator author a correct shared-buffer story on day one, before Zicbom is implemented or trusted. Keep it afterwards — it is the fastest way to bisect a suspected coherence bug.

Figure 7 — Physical memory map and attributes.



Both checks run **in parallel with D\$ tag lookup**, not after it — the result is needed in the same cycle to decide whether the access may proceed and whether it may be cached.

The rules that matter for accelerators (normative, `INTERFACES.md` P1-P5):

- **P1** — a non-idempotent region is never speculatively accessed, prefetched or replayed. This is a structural exclusion in the D\$, not a convention.
- **P2** — `order: strong` regions are accessed in program order without a fence. This is what makes writing an accelerator’s descriptor block and then its `CTRL.start` register safe.
- **P3** — an alignment or width violation raises an access fault, never a silent truncation.

The same check unit is instantiated on the coprocessor’s memory port. Otherwise a coprocessor could bypass PMP entirely and the platform’s PMP story would be hollow. PMP configuration stays core-owned; the coprocessor port gets a read-only copy of the config.

12. Performance counters

Without these, no accelerator claim on this platform is measurable, and §31-34 do not work.

`mcycle`, `minstret`, `mcountinhibit`, and `mhpmcounter3-15` with programmable `mhpmevent3-15`.

Group	Events
Frontend	<code>icache_miss</code> , <code>icache_stall_cycles</code> , <code>branch_taken</code> , <code>branch_mispredict</code> , <code>fetch_stall</code>
Backend	<code>load_use_stall</code> , <code>csr_serialise_stall</code> , <code>cb_full_stall</code> , <code>div_busy_cycles</code>
Memory	<code>dcache_miss</code> , <code>dcache_writeback</code> , <code>store_buffer_full</code> , <code>dtlb_miss</code> , <code>itlb_miss</code>
Coprocessor	<code>mxif_offload_count</code> , <code>mxif_issue_stall_cycles</code> , <code>mxif_busy_cycles</code> , <code>mxif_wb_stall_cycles</code>
Fabric	<code>axi_read_beats</code> , <code>axi_write_beats</code> , <code>axi_read_latency_sum</code> , <code>axi_arb_stall_cycles</code>
Traps	<code>exception_taken</code> , <code>interrupt_taken</code>

The coprocessor and fabric groups are the ones that make accelerator research possible. They let a student answer *why* a speedup was disappointing:

Symptom	Attribution
<code>mxif_busy_cycles</code> low, total cycles high	the accelerator is idle — the problem is software or the offload path, not the datapath
<code>mxif_issue_stall_cycles</code> high	the core cannot feed the accelerator — offload overhead dominates; consider loose coupling
<code>axi_read_latency_sum</code> / <code>axi_read_beats</code> high	memory-bound — the accelerator is starved; look at tiling and double buffering (§22)
<code>axi_arb_stall_cycles</code> high	bus contention with another master — check the bandwidth budget (§18.3)
<code>accel_busy</code> high, speedup still low	genuinely compute-bound — the datapath is the honest bottleneck

That last row is the only one where the accelerator’s design is the answer. **A thesis that cannot rule out the other four rows has not measured anything.**

12.1 Software interface

```
#include <s1_perf.h>

perf_config(PERF_CTR0, EV_MXIF_BUSY_CYCLES);
perf_config(PERF_CTR1, EV_AXI_READ_BEATS);

perf_region_t r;
PERF_BEGIN(&r);
    conv2d_accel(input, weights, output);
PERF_END(&r);

perf_report(&r); // cycles, instret, and every configured counter, to UART or semihosting
```

If measurement is not this easy, it will not be done, and every result in the lab will be incomparable. Ship `libs1_perf` in Phase 2, not Phase 6.

13. Debug

RISC-V External Debug Support via `pulp-platform/riscv-dbg`. Core-side obligations are listed normatively in `INTERFACES.md` §7. Summary of what touches the pipeline and therefore must be designed in Phase 1 even though the DM lands in Phase 3:

- `DebugMode` in the privilege FSM (§10.1), `dcsr/dpc/dscratch0/dscratch1`.
- Halt request handling at the retire point; **entry waits for `x_idle`**.
- Single-step via `dcsr.step`.
- Trigger module: `tselect`, `tdata1/2/3`, ≥ 2 instruction-address triggers.
- Abstract command support for register and memory access — this is what makes `load` over JTAG work.

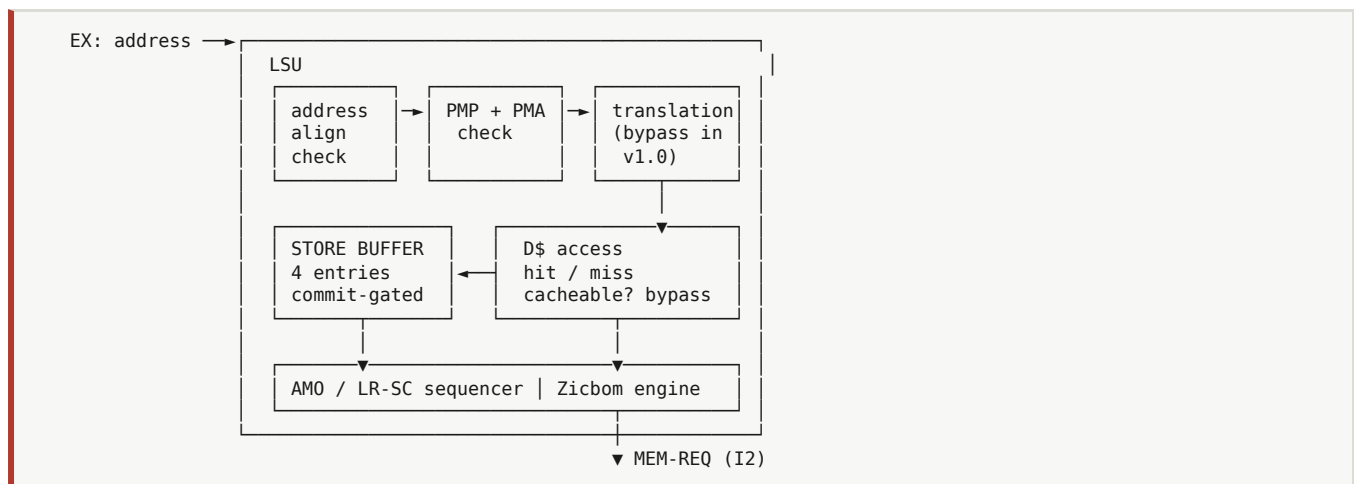
The user-visible result, which is FR-9 and the thing that unblocks the whole lab:

```
$ openocd -f boards/kc705/openocd.cfg &  
$ riscv64-unknown-elf-gdb my_app.elf  
(gdb) target extended-remote :3333  
(gdb) load  
(gdb) break conv2d  
(gdb) continue
```

No resynthesis, ever, for a software change.

Part III — Memory

14. Load/store unit



The translation stage is present from v1.0 as a bypass. This costs one mux and one pipeline boundary today. Adding it in Phase 5 would mean re-timing the entire load path with a working Linux port on top — the worst possible moment.

Store buffer entries are allocated at MEM and committed at retire. A store never becomes visible before its instruction retires. This is what makes precise exceptions work on the store path.

15. Caches

Direct-mapped or 2-way, parameterised, blocking, physically indexed and physically tagged (trivially true in v1.0 with translation bypassed; still true in Phase 5 because the index fits within the page offset at these sizes).

D\$ organisation — 8-16 KB, 2-way, 64 B lines, write-back

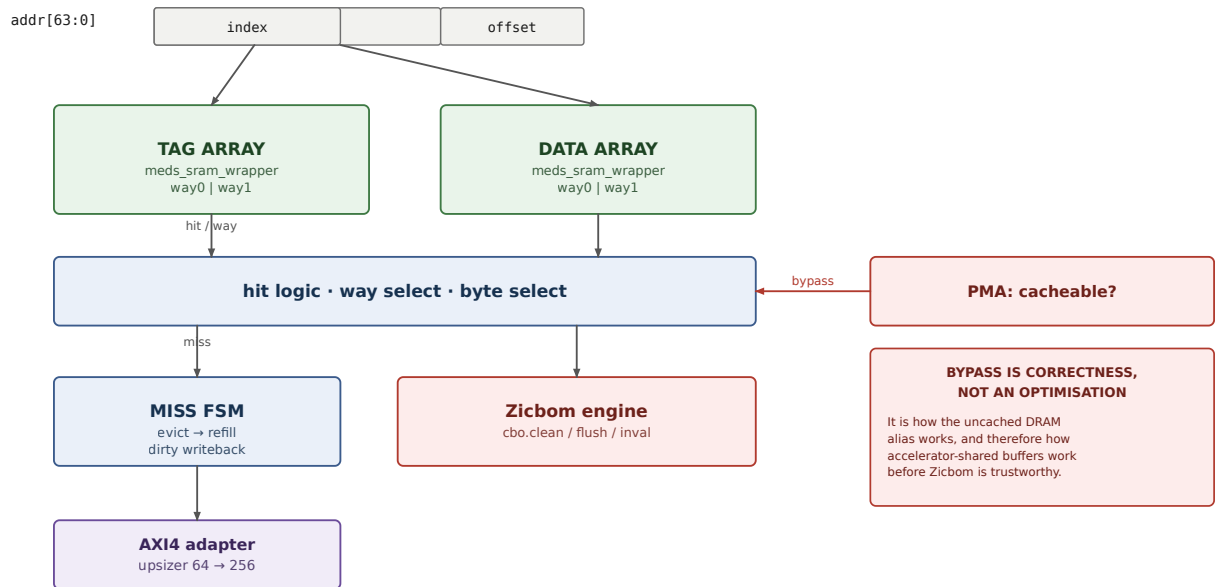
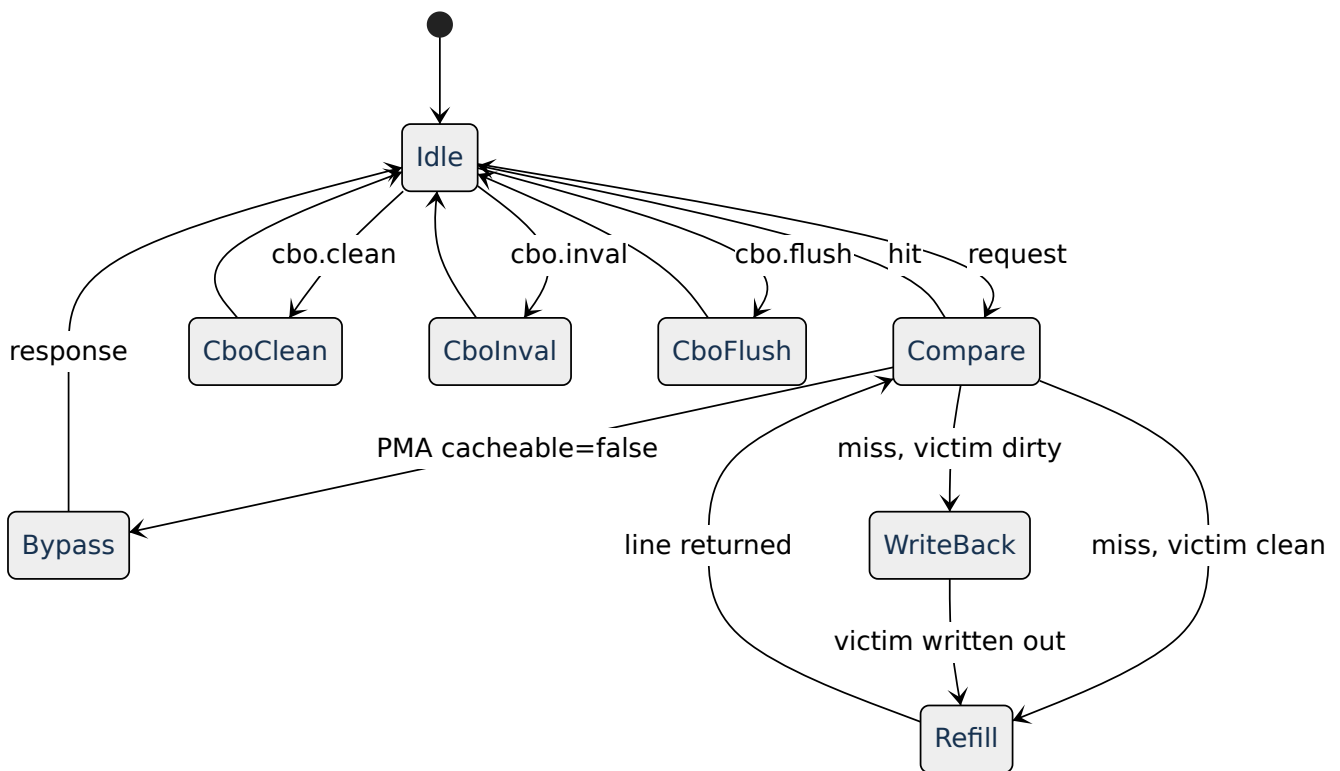


Figure 8 — D\$ organisation.



Bypass on `cacheable: false` is not an optimisation — it is a correctness requirement. It is how the uncached DRAM alias (§18.4) works, and therefore how accelerator-shared buffers work before Zicbom is trustworthy.

16. MMU and page-table walker

Sv39, three-level, 32-entry fully-associative TLB per port. **Implemented in Phase 5; architected in Phase 1.**

Page-table walker — the second port that must exist in v1.0

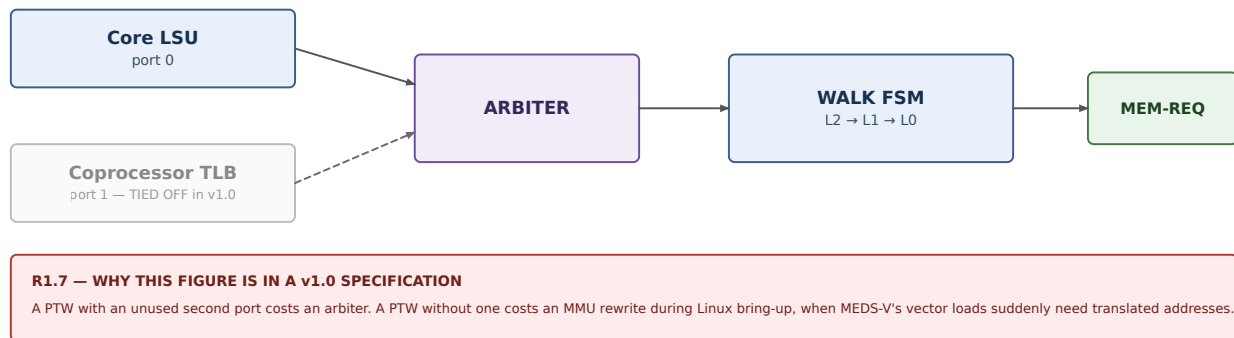


Figure 9 — Page-table walker with the reserved second port.

Port 1 is the entire reason this section exists in a v1.0 specification. A PTW with an unused second port costs an arbiter. A PTW without one costs an MMU rewrite during Linux bring-up, when MEDS-V's vector loads suddenly need translated addresses ([ADDENDUM.md](#) A3).

Coprocessor memory model, restated: **M2 in v1.0** (physical addresses, M-mode, non-cacheable window or [Zicbom](#)-managed), **M3 in Phase 5** (coprocessor-side TLB filled through PTW port 1).

17. SRAM wrapper

Every array in the design — register file, cache tags, cache data, TLB, VRF, deep FIFOs — goes through `meds_sram_wrapper` ([INTERFACES.md](#) §8). No exceptions, enforced at review.

Read latency is **one cycle, registered output, everywhere**. A mixed-latency memory system is where timing closure and verification both go to die.

Implementations selected by parameter: behavioural (simulation), FPGA BRAM, ASIC macro. This one rule is what keeps a tape-out possible without a rewrite.

Part IV — Integration

This part is the platform's reason for existing. If a reader has time for only one section, it is this one.

18. Bus fabric

AXI4 fabric topology and bandwidth budget

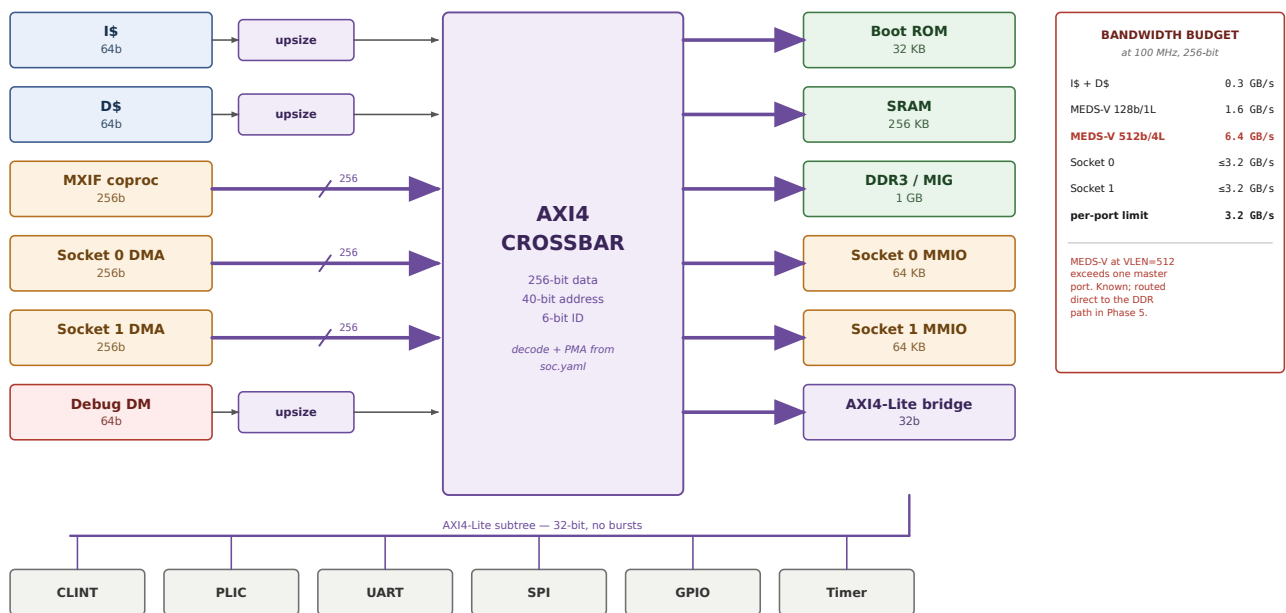
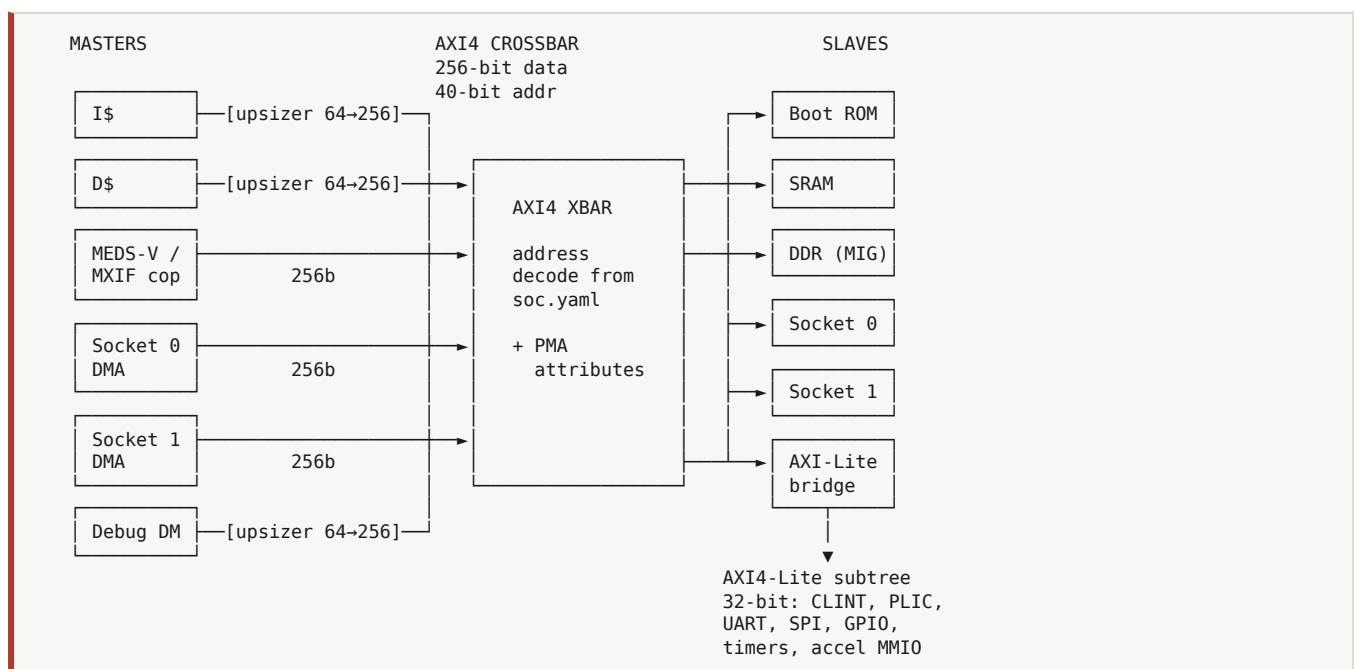


Figure 10 — AXI4 fabric topology and the bandwidth budget.

18.1 Topology



18.2 Parameters

	Backbone	Peripheral subtree
Protocol	AXI4	AXI4-Lite
Data width	256 bit (S1-AI) / 64 bit (S1-Base)	32 bit
Address width	40 bit	40 bit
ID width	6 bit	—
Bursts	INCR $\leq$ 16 beats; WRAP for refill	none
Implementation	pulp-platform/axi	pulp-platform/axi lite

18.3 The bandwidth budget

Every accelerator declares its demand against this table before it is accepted into the platform. An accelerator that exceeds its budget is a design review item, not a merge.

At 100 MHz, 256-bit backbone: 3.2 GB/s per master port.

Master	Budget	Basis
I\$ + D\$ refill	0.3 GB/s	64-bit port via upsizer
MXIF coprocessor (MEDS-V, VLEN=128, 1 lane)	1.6 GB/s	16 B/cycle unit-stride
MXIF coprocessor (MEDS-V, VLEN=512, 4 lanes)	6.4 GB/s	64 B/cycle — exceeds one port; needs the DDR path directly
Socket 0 DMA	$\leq$ 3.2 GB/s	declared per accelerator
Socket 1 DMA	$\leq$ 3.2 GB/s	declared per accelerator
Debug	negligible	

The MEDS-V row at VLEN=512 is a known future problem and is recorded here rather than discovered later. It is exactly the finding MEDS-V’s own book anticipates in Ch 16.2 (“the VLEN-wide idealised port does not exist on real hardware”).

(DDR3 on KC705 supplies substantially more than the backbone can carry, so the backbone is the binding constraint, not the DRAM. Confirm the MIG user-interface width and clock for your specific part during Phase 2 bring-up and record the measurement here.)

18.4 The uncached DRAM alias

DRAM is mapped twice:

Alias	Base	Cacheable	Use
dram	0x8000_0000	yes	ordinary program data
dram_uncached	0x1_0000_0000	no	accelerator-shared buffers

Costs one address-decode bit. Gives every accelerator author a correct shared-buffer story on day one, before Zicbom is implemented or trusted. Keep it after Zicbom lands — it remains the fastest way to bisect a suspected coherence bug.

19. Tight coupling — MXIF

Use when: the unit operates on register-file values, has short latency, and benefits from being in the instruction stream. MEDS-V, custom ALU ops, crypto rounds, activation functions, posit units.

Full normative specification: [INTERFACES.md §1](#). This section is the explanatory view.

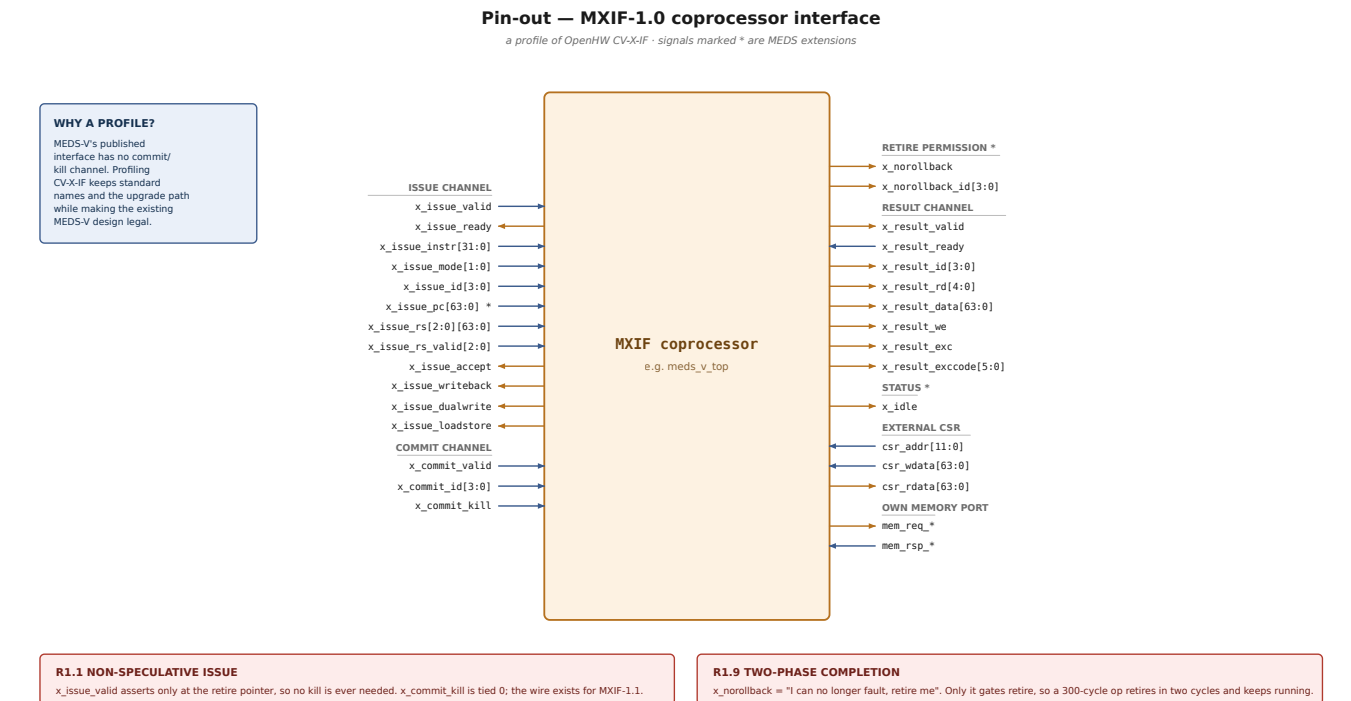
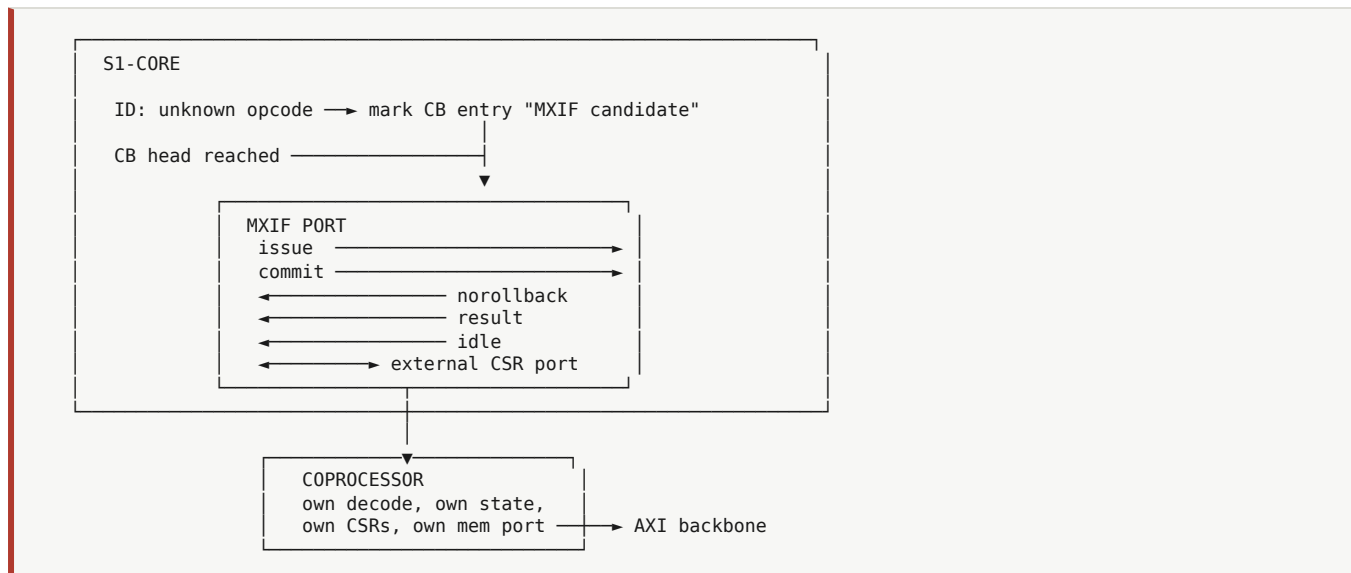


Figure 11 — MXIF-1.0 coprocessor interface pin-out.

19.1 Structure



19.2 Timing — a vector load, the decoupled case

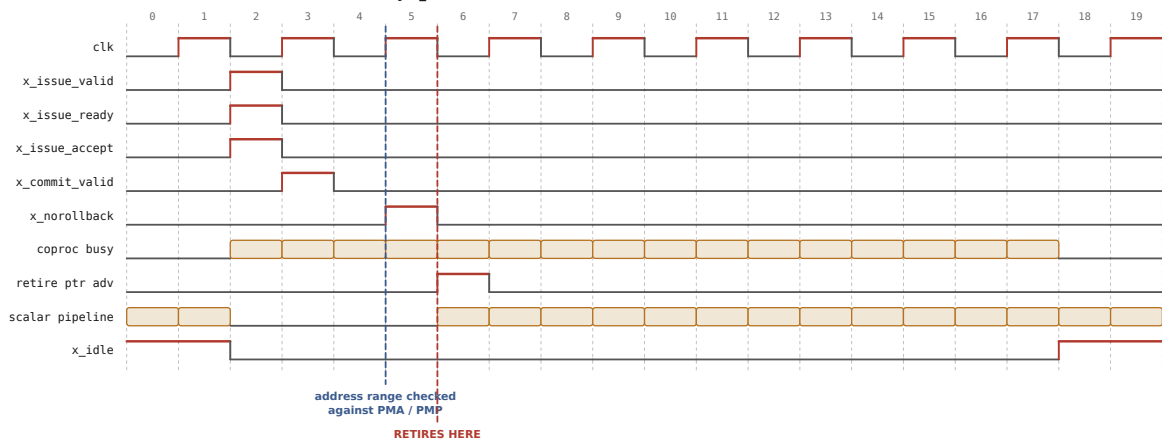
Trace `vle32.v v1, (a0)` through the interface. The instruction is offloaded at the retire pointer (cycle 2), the coprocessor accepts, and the commit channel fires the following cycle. The VLSU then computes the `v1`-derived byte range and checks it against PMA and PMP; when that passes it asserts `x_norollback` (cycle 5).

Because this instruction writes no scalar register, §9.2’s retire rule needs only `x_norollback` — so it **retires at cycle 6 while the coprocessor is still fetching data**, and the scalar pipeline runs unrelated work from cycle 6 to cycle 18.

That window is the decoupling. It is the behaviour MEDS-V book §8.6 calls “the whole argument for decoupling”, and R1.9 is what makes it legal: the instruction is architecturally invisible once retired, so it must have already proved it cannot fault.

MXIF two-phase completion — why decoupling survives in-order retire

vle32.v v1,(a0) — no scalar writeback, so retire needs only `x_norollback`



The vector load is architecturally retired at cycle 6 and is still fetching data until cycle 18.

The scalar pipeline runs unrelated work throughout — this is the entire argument for decoupling, and it is only legal because of R1.9.

vsetvli a0, a1, e32, m1 — writes a scalar register, so retire needs both

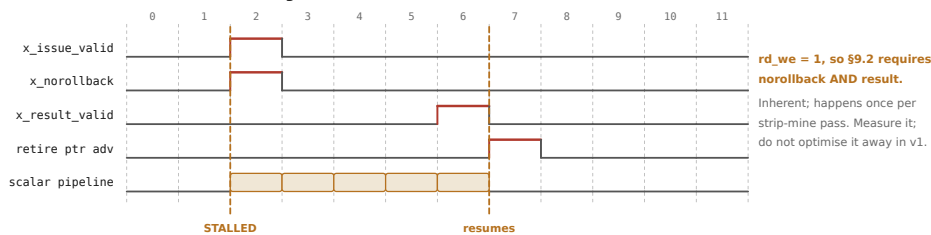


Figure 12 — MXIF two-phase completion timing.

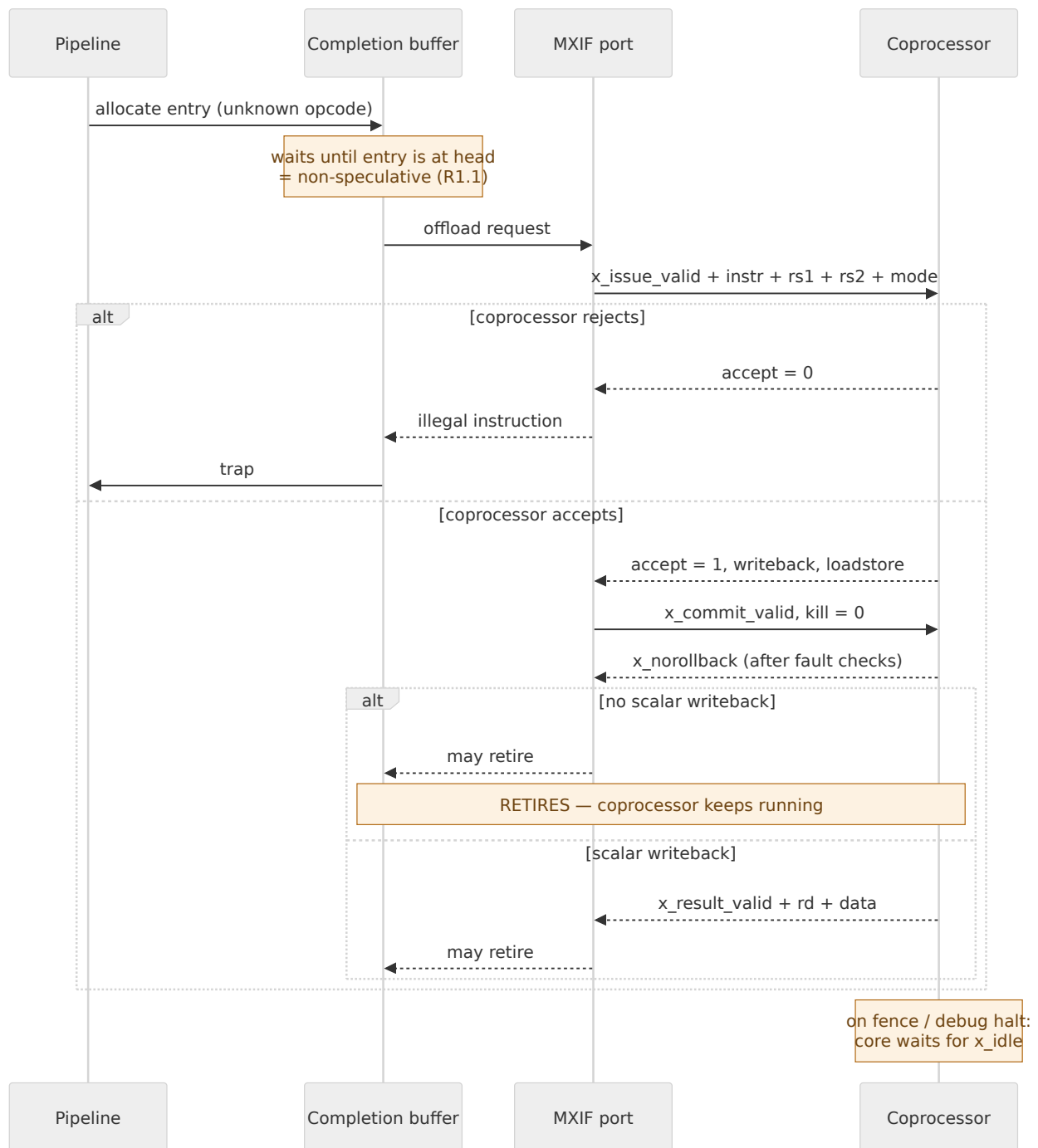
19.3 Timing — `vsetvli`, the synchronous case

`vsetvli` returns the new `v1` to a scalar register, so `x_issue_writeback` is asserted and §9.2 requires **both** `x_norollback` and `x_result_valid` before it may retire. The core stalls from issue until the result arrives.

This is inherent, not a defect: the *next* scalar instruction in a strip-mine loop almost always consumes `v1` to compute a pointer bump, so the loop synchronises once per pass. MEDS-V’s book already warns about it. **Measure it; do not try to optimise it away in v1.**

Both cases are shown in the timing figure above.

19.4 Handshake sequence



19.5 Adding a custom instruction — the seven-item checklist

Nothing merges without all seven. This is what makes MXIF an extension *mechanism* rather than a reserved opcode.

#	Artefact	Location
1	riscv-opcodes-format encoding	extensions/<name>/<name>.opcodes
2	Opcode-space allocation entry	extensions/REGISTRY.md
3	Golden-reference model (Spike plugin or Sail patch)	extensions/<name>/model/
4	Toolchain path (.insn macro + inline-asm header minimum)	extensions/<name>/sw/
5	RISCOF-format tests against the model from (3)	extensions/<name>/tests/
6	CSR allocation entry, if any (prefer standard CSRs over custom)	extensions/REGISTRY.md
7	README.md : semantics, latency, exceptions, fence interaction	extensions/<name>/

20. Loose coupling — the accelerator socket

Use when: the unit has its own memory access pattern, high arithmetic intensity, and runs for thousands of cycles per invocation. Systolic arrays, convolution engines, FFT blocks, crypto bulk engines, DSP pipelines. **This is the mechanism most DL/ML thesis projects should use.**

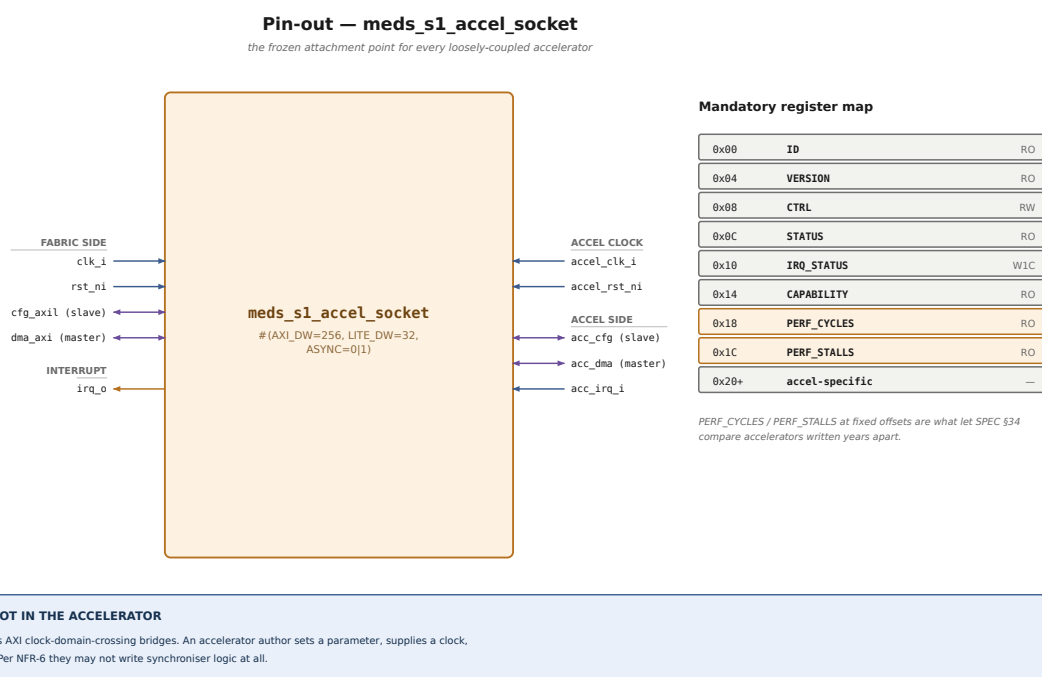
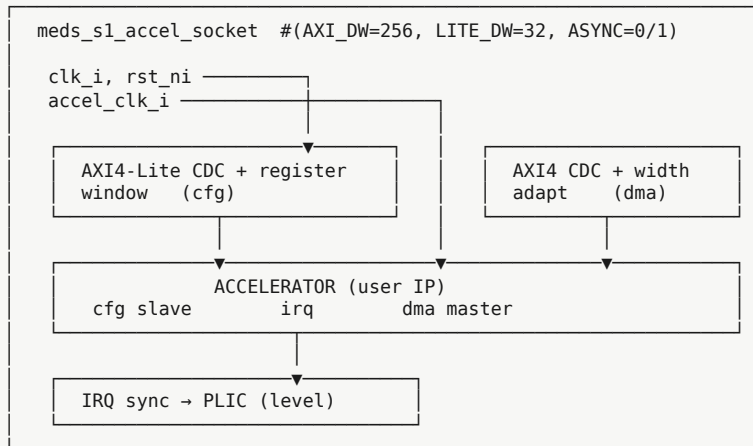


Figure 13 — Accelerator socket pin-out and mandatory register map.

20.1 Structure



Clock-domain crossing lives in the socket, never in the accelerator. An accelerator author sets `ASYNC=1`, supplies a clock, and writes purely synchronous logic. Per NFR-6, they are not permitted to write synchroniser logic themselves.

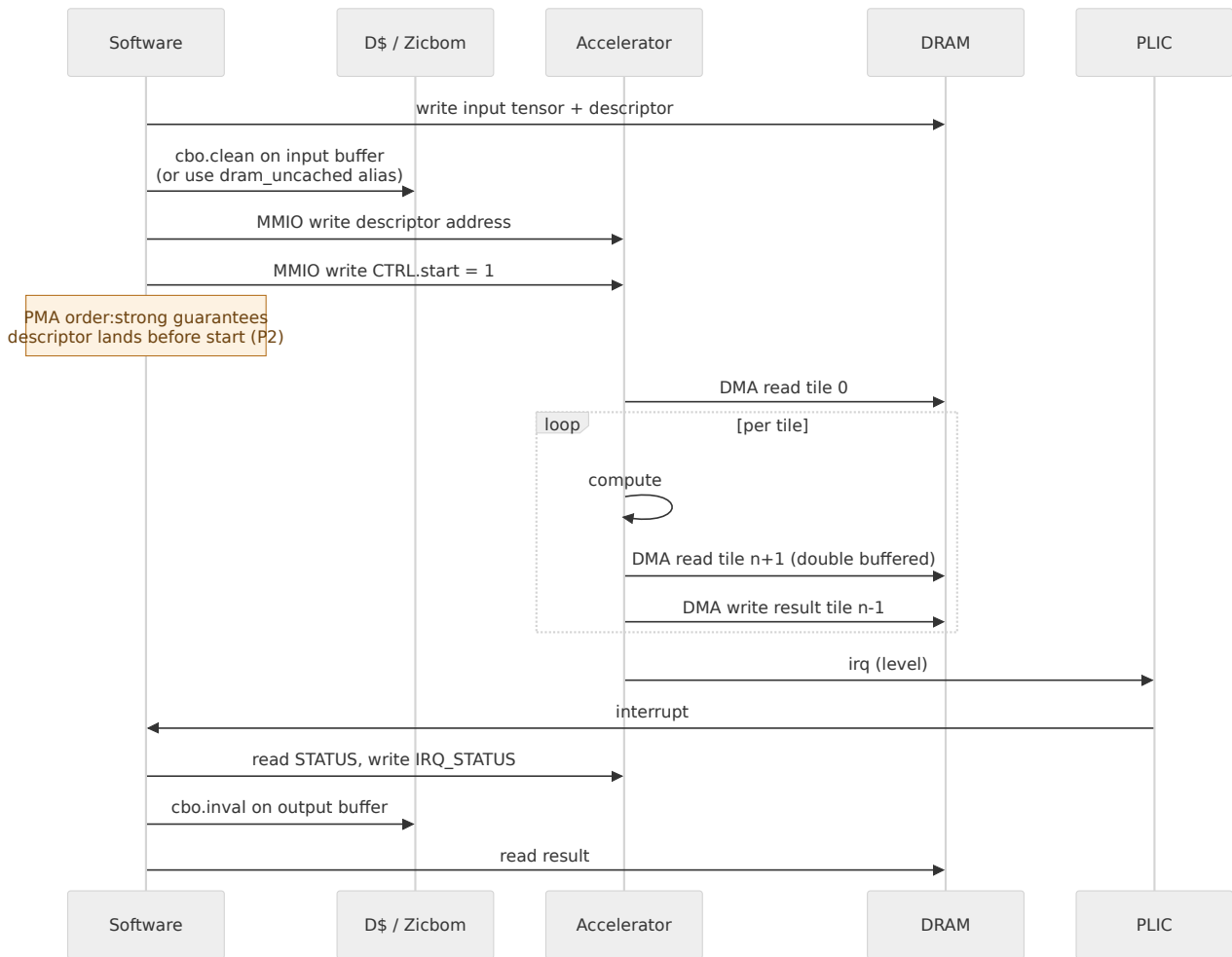
20.2 Mandatory register map

Every accelerator's `cfg` window begins with this block. It costs nothing and it means the BSP can enumerate and identify every accelerator on the bus without accelerator-specific code.

Offset	Name	Access	Contents
0x00	ID	RO	allocated in extensions/REGISTRY.md
0x04	VERSION	RO	{major[15:0], minor[15:0]}
0x08	CTRL	RW	[0] start [1] abort [2] irq_en
0x0C	STATUS	RO	[0] busy [1] done [2] error [7:4] errcode
0x10	IRQ_STATUS	W1C	write-1-to-clear
0x14	CAPABILITY	RO	feature bits, accelerator-defined
0x18	PERF_CYCLES	RO	accelerator-local busy cycle count
0x1C	PERF_STALLS	RO	accelerator-local memory stall count
0x20+	—		accelerator-specific

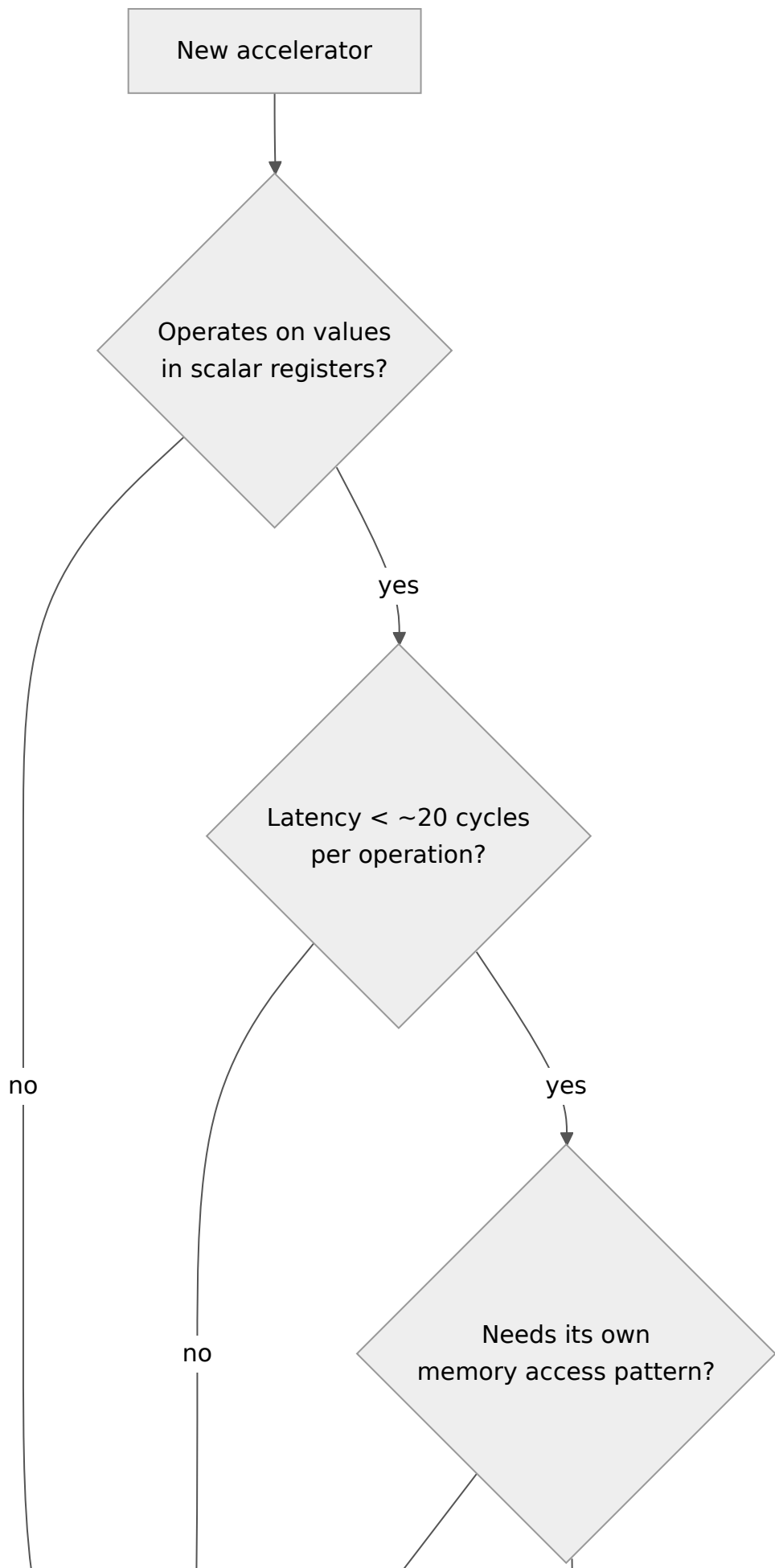
`PERF_CYCLES` and `PERF_STALLS` at fixed offsets are what let §34's comparison methodology work across accelerators written by different students in different years.

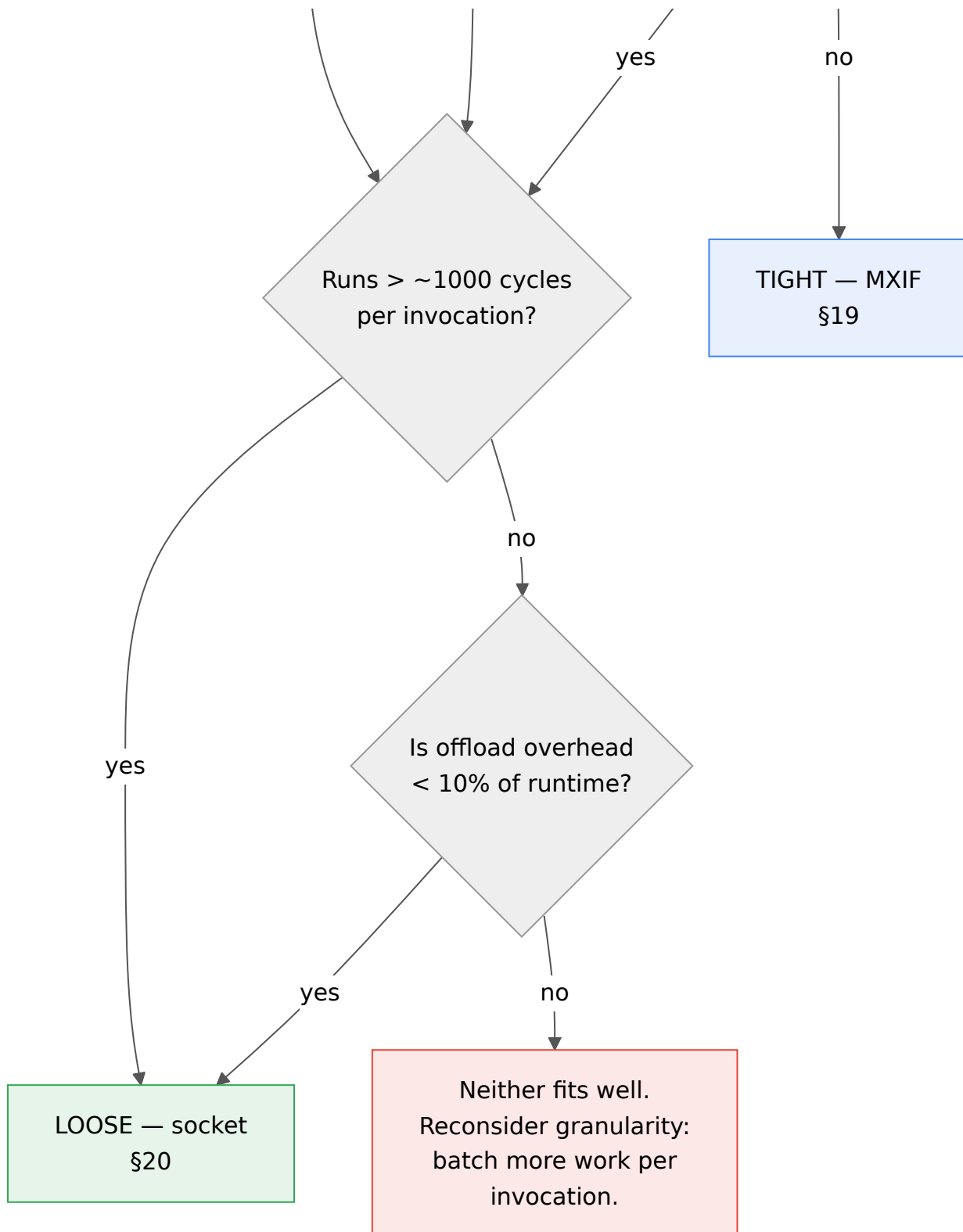
20.3 Invocation flow



Steps 2 and 10 — the cache maintenance — are the ones students forget, and the symptom is stale data that looks like an accelerator bug. **The driver template ships with them already written.**

21. Choosing a coupling mechanism





	Tight (MXIF)	Loose (socket)
Invoked by	an instruction	MMIO write
Operands	scalar registers, forwarded	memory, via DMA
Invocation overhead	~2-5 cycles	~50-200 cycles (MMIO + IRQ)
Good granularity	1-100 cycles of work	1000+ cycles of work
Memory access	its own port, or none	its own DMA master
Needs toolchain work	yes — 7-item checklist	no — it's just a driver

	Tight (MXIF)	Loose (socket)
Concurrency with scalar	yes, after <code>x_norollback</code>	yes, fully
Typical examples	MEDS-V, crypto rounds, activations, posit ALU	conv engines, systolic arrays, FFT, DL accelerators
Recommended for most DL/ML theses		✓

The common mistake: attaching a convolution engine over MXIF because instructions feel more elegant. A convolution runs for thousands of cycles and streams megabytes; it does not want to be an instruction. Use the socket. Conversely, attaching a single-cycle activation function over the socket makes a 3-cycle operation cost 200 cycles of MMIO overhead.

22. Data movement patterns for ML workloads

For edge-AI accelerators the bottleneck is almost never arithmetic. This section exists because most first accelerator designs are compute-optimal and memory-naive, and the platform should teach the difference before the student's third month.

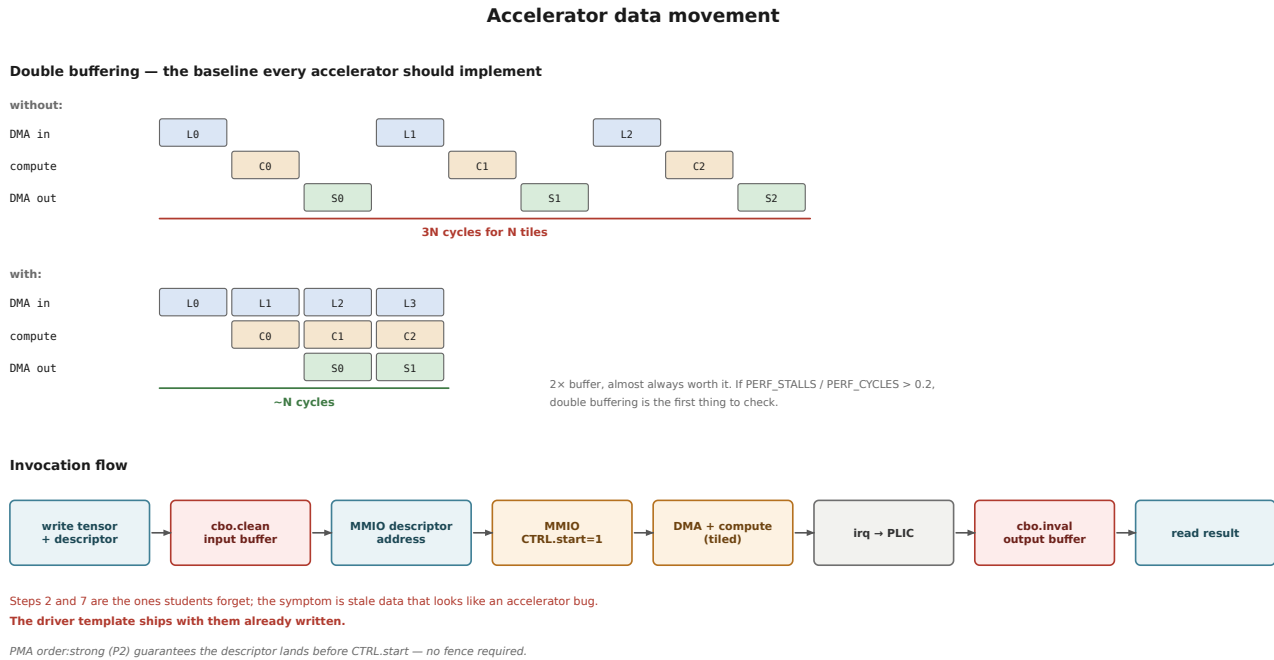


Figure 14 — Accelerator data movement: double buffering and invocation flow.

22.1 Double buffering — the baseline every accelerator should implement

Without double buffering:

```

|load T0| |comp T0| |store T0| |load T1| |comp T1| |store T1|
|----- 3N cycles for N tiles -----|

```

With double buffering:

```

|load T0| |load T1| |load T2| |load T3|
|comp T0| |comp T1| |comp T2|
|store T0| |store T1|
|----- ~N cycles, if load ≈ comp ≈ store -----|

```

Requires 2x the on-chip buffer. Almost always worth it. If $\text{PERF_STALLS} / \text{PERF_CYCLES}$ exceeds ~ 0.2 , double buffering is the first thing to check.

22.2 Tiling and the roofline

The generic edge-AI convolution tiling structure, and the parameters an accelerator must expose so a thesis can sweep them:

```

for (oc_tile : output channels)      — T_oc
  for (oh_tile : output height)      — T_oh
    for (ow_tile : output width)     — T_ow
      load weights[oc_tile][*]      — reuse across oh,ow
      for (ic_tile : input channels) — T_ic
        load ifmap tile             — reuse across oc
        compute
        store ofmap tile

```

Arithmetic intensity, and therefore whether the design is memory- or compute-bound, is $T_{oc} \times T_{ic} \times T_{oh} \times T_{ow}$ MACs per byte moved. **Make the tile sizes runtime-configurable registers, not synthesis parameters.** A thesis that can sweep tiling in software produces a roofline plot; one that must resynthesise for each point produces three data points.

22.3 Buffer placement decision

Buffer	Where	Why
Weights (small, reused)	accelerator-local SRAM	maximum reuse, no bus traffic
Activations (streamed)	DMA from <code>dram_uncached</code> , double buffered	too large to hold
Output tiles	accelerator-local, then DMA out	write coalescing
Descriptors, control	<code>dram_uncached</code> or SRAM	tiny, latency-sensitive
Anything the CPU also touches	<code>dram_uncached</code> , or <code>dram</code> + <code>Zicbom</code>	correctness (§18.4)

22.4 Sizing the interface before building the datapath

Before writing a line of accelerator RTL, compute:

```
required bandwidth (B/s) = (bytes moved per inference) / (target latency)
available bandwidth (B/s) = backbone_width/8 × f_clk × achievable_efficiency (~0.7)
```

If required > available, **the datapath does not matter yet** — fix the data movement first. This calculation is a required section of every accelerator proposal in the lab (§33).

23. Worked example — attaching a CNN accelerator

An end-to-end walkthrough, kept concrete because this is the path most MS students will take.

23.1 What the student writes

```
accelerators/meds-x-conv/
├── rtl/
│   ├── conv_top.sv          ← cfg slave, dma master, irq (the only required file)
│   ├── conv_pe_array.sv
│   ├── conv_ctrl.sv
│   └── conv_linebuf.sv
├── sw/
│   ├── conv_driver.c        ← from the template
│   └── conv_driver.h
├── verif/
│   ├── tb_conv_socket.sv    ← socket conformance TB, provided
│   └── tb_conv_unit.sv
├── docs/README.md           ← incl. the §22.4 bandwidth calculation
└── accel.yaml               ← ID, version, bandwidth demand, register map
```

Everything else — crossbar wiring, address decode, IRQ routing, C header generation, device tree, linker script — is generated.

23.2 Attaching it

```
# soc.yaml
sockets:
  - id: 0
    ip: meds-x-conv
    base: 0x2000_0000
    size: 64K
    irq: 16
    async_clk: true
    clk_mhz: 150
    bandwidth_gbps: 2.4      # checked against the $18.3 budget by the generator
```

```
$ make soc CONFIG=s1_ai
$ make verilate && make test-accel      # runs in simulation first
$ make bitstream BOARD=kc705
$ make run BOARD=kc705 PROG=apps/ecg_cnn.elf
```

23.3 The driver, from the template

```
#include <s1_accel.h>
#include "conv_driver.h"

int conv2d_accel(const int8_t *ifmap, const int8_t *weights,
                 int8_t *ofmap, const conv_cfg_t *cfg)
{
    accel_t *a = accel_open(ACCEL_ID_CONV);      // enumerates by ID register
    if (!a) return -ENODEV;

    conv_desc_t d = {
        .ifmap = (uintptr_t)ifmap, .weights = (uintptr_t)weights,
        .ofmap = (uintptr_t)ofmap,
        .t_oc = cfg->t_oc, .t_ic = cfg->t_ic,      // runtime tiling -- see 22.2
        .t_oh = cfg->t_oh, .t_ow = cfg->t_ow,
    };

    accel_cache_clean(ifmap, cfg->ifmap_bytes);    // §20.3 step 2
    accel_cache_clean(weights, cfg->weight_bytes);

    accel_write_desc(a, &d);
    accel_start(a);                               // CTRL.start, PMA order:strong
    accel_wait_irq(a);                            // WFI until PLIC fires

    accel_cache_invalidate(ofmap, cfg->ofmap_bytes); // §20.3 step 10
    return accel_status(a);
}
```

23.4 Measuring it

```
perf_config(PERF_CTR0, EV_AXI_READ_BEATS);
perf_config(PERF_CTR1, EV_AXI_ARB_STALL_CYCLES);

PERF_BEGIN(&r);
    conv2d_accel(ifmap, weights, ofmap, &cfg);
PERF_END(&r);

perf_report(&r);
accel_perf_report(a);      // reads PERF_CYCLES / PERF_STALLS from the socket registers
```

Output, in the format §34 requires for comparability:

```
=== MEDS-S1 perf report ===
config       : s1_ai @ 100 MHz, KC705
workload     : ecg_cnn, 1 inference
cycles       : 912,431      (9.12 ms)
instret      : 118,204
axi_read_beats : 44,102      (1.41 MB)
axi_arb_stall : 21,880       (2.4%)
accel_busy    : 712,004      (78.0%)
accel_stall    : 128,410      (14.1% -- memory bound)
scalar baseline : 12,304,551 cycles
speedup      : 13.5x
```


The `accel_stall` line is what turns this from a number into a finding.

Part V — SoC and software

24. Peripherals and interrupts

Peripheral	Bus	Source	Notes
CLINT	AXI4-Lite	reuse	<code>msip</code> , <code>mtime</code> , <code>mtimecmp</code> — required for Linux
PLIC	AXI4-Lite	reuse	priority, threshold, claim/complete
UART	AXI4-Lite	reuse (OpenTitan/PULP)	16550-compatible preferred; console + second-stage load
SPI	AXI4-Lite	reuse	SD card, QSPI flash
GPIO	AXI4-Lite	reuse	LEDs, switches, bring-up
Timer	AXI4-Lite	reuse	general-purpose
Accelerator MMIO	AXI4-Lite	generated	one window per socket

Interrupt allocation policy — IDs come from `soc.yaml`, generated into the device tree and C headers. Never hand-assigned in two places.

ID range	Reserved for
0	no interrupt (PLIC convention)
1-15	platform peripherals
16-31	accelerator sockets
32+	expansion

All PLIC sources are **level-sensitive**. Edge-triggered interrupts across a clock-domain crossing are a bug generator, and §20.1 puts a CDC in every asynchronous socket.

25. Clock, reset and power

```
board_top.sv (the only board-aware file with logic)

ext_clk → PLL/MMCM → clk_core (100 MHz)
                  → clk_ddr  (MIG UI)
                  → clk_accel0 (150 MHz, optional)
                  → clk_accel1 (optional)

ext_rst_n → reset synchroniser per domain
          async assert, sync de-assert, active low
```

Domain crossings, all confined to named modules:

```
clk_core ↔ clk_ddr      : AXI CDC in the MIG adapter
clk_core ↔ clk_accelN   : AXI CDC inside med_s1_accel_socket (§20.1)
clk_core ↔ jtag_tck     : inside riscv-dbg DTM
```

Reset policy, stated once and enforced: asynchronous assert, synchronous de-assert, active-low (`rst_ni`). One reset per clock domain. No local resets, no reset generation inside leaf modules.

Power is out of scope for v1.0 measurement, but two cheap hooks go in now because they cannot be retrofitted cheaply: clock-gating enables on the accelerator sockets and on MEDS-V, and a documented `wfi` behaviour that actually gates the core clock. A future energy-efficiency thesis needs both.

26. The SoC generator

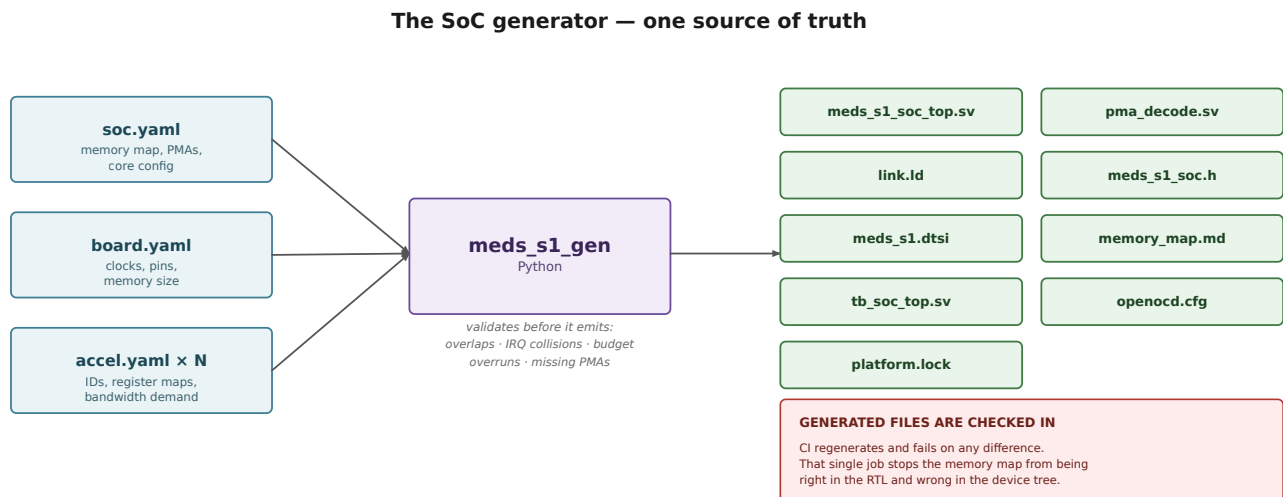


Figure 15 — *The SoC generator: one source of truth.*

Generation rules:

- The generator **validates before it emits**: overlapping regions, IRQ collisions, bandwidth budget overruns (§18.3), missing PMA attributes, and unallocated accelerator IDs are all errors.
- Generated files carry a “DO NOT EDIT — generated from soc.yaml” header and are **checked in**, so a diff shows what a config change actually did. CI regenerates and fails on any difference.
- Golden-file tests for every named config in §5.3.

The acceptance test for the generator is NFR-9: a contributor who has done only the workshop adds a peripheral by editing YAML and sees it from C, following the docs alone, in under a day.

27. Software stack

Software stack and the host toolchain



Semihosting is the single feature that will make students believe the platform is real:

it is how a 10 MB weight file reaches the FPGA without an SD card, and how `printf` works before UART is trusted.

The Verilator target is not a lesser mode — it is where most development happens, and it needs no board.

Figure 16 — Software stack and host toolchain.

27.1 The `make run` contract

```
$ make run BOARD=verilator PROG=apps/ecg_cnn.elf # no hardware needed
$ make run BOARD=kc705    PROG=apps/ecg_cnn.elf # loads over JTAG, no resynthesis
$ make debug BOARD=kc705  PROG=apps/ecg_cnn.elf # same, drops into gdb
```

Identical invocation across boards. The Verilator target is not a lesser mode — it is where most development happens (§29.3), and it doubles the lab’s effective capacity because it needs no board.

27.2 Semihosting

`open`, `read`, `write`, `close`, `lseek` proxied to the host filesystem through the Debug Module. This is how a 10 MB weight file reaches the FPGA without an SD card, and how `printf` works before UART is trusted. **It is the single feature that will make students believe the platform is real.**

Part VI — Verification and implementation

28. Verification architecture

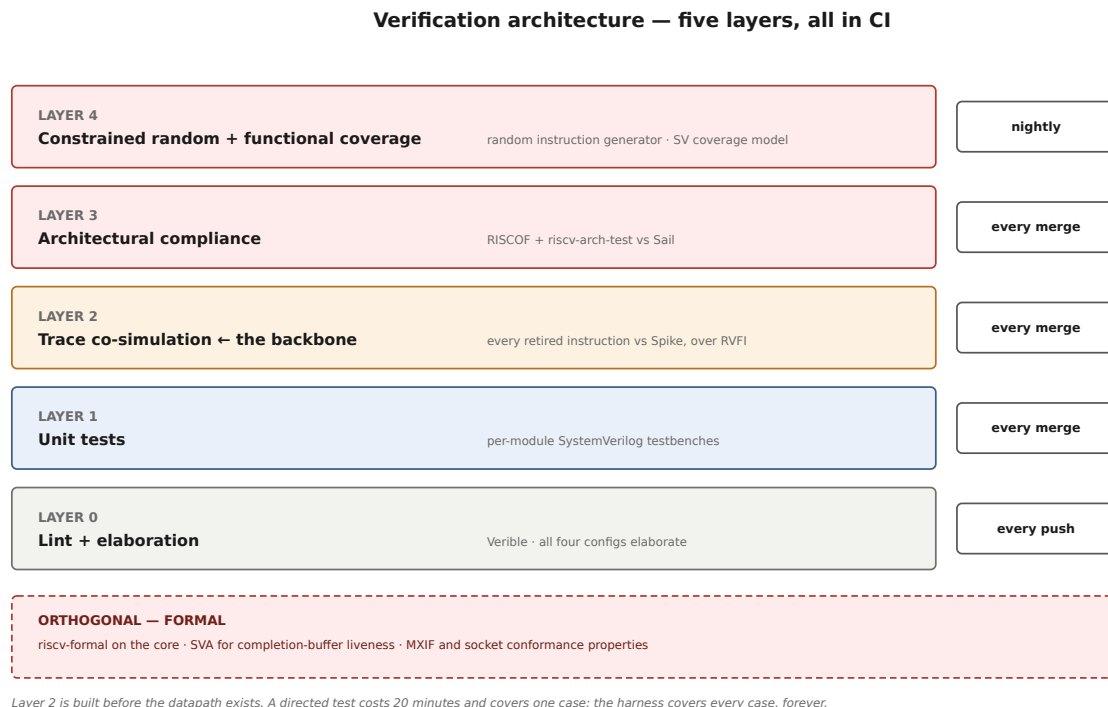


Figure 17 — Verification architecture: five layers, all in CI.

Layer 2 is built before the datapath exists. MEDS-V’s book puts co-simulation at M2 for exactly this reason, and MEDS-S1 follows: the RVFI port and the Spike comparison harness are Phase-1 deliverables, not Phase-4.

28.1 Why co-simulation dominates

A directed test costs ~20 minutes and covers one case. A co-simulation harness costs ~3 days and covers every case any program exercises, forever. Without it the only signal is “the final answer is wrong”; with it the signal is “instruction 4127, PC 0x800001a4, add wrote x5=0x..., expected 0x...”.

28.2 The RVFI contract

RVFI at retire, one channel per retiring instruction, plus the MEDS `rvfi_v_*` group for vector state (INTERFACES.md §5). Byte-granular `rvfi_v_vd_wmask` is what lets the checker verify MEDS-V’s undisturbed-tail policy, which is invisible to a checker that compares only final register contents.

28.3 Conformance testbenches — the reusable assets

Two testbenches that outlive every accelerator:

- `verif/conformance/tb_mxif_conformance.sv` — drives a coprocessor through every MXIF handshake case, including rejection, early `norollback`, late result, fault before `norollback`, and `x_idle` behaviour under `fence`. Checks the liveness properties of §9.4.
- `verif/conformance/tb_socket_conformance.sv` — drives an accelerator through the mandatory register map, IRQ assert/clear, abort, error reporting, and DMA to legal and illegal regions.

Every new accelerator must pass the relevant conformance TB before it is accepted. This is the mechanism that stops accelerator #7 from breaking assumptions accelerator #3 relied on.

28.4 Coverage model

Functional coverage of the implemented ISA: instruction × operand class × hazard class × privilege × trap. Plus interface coverage: MXIF handshake cases, socket transaction types, PMA region × access type crosses. Published per release.

28.5 CI policy

Trigger	Runs	Budget
every push	lint, elaborate 4 configs	< 5 min
every PR	+ unit tests, arch-tests, co-simulation, benchmarks	< 20 min (NFR-3)
nightly	+ random regression, formal, synthesis, timing, area	< 4 h
release	+ all configs, full arch-test, coverage, evidence bundle	—

Nothing merges red. A performance regression > 3% blocks merge without written justification (NFR-10).

28.6 The evidence bundle

Every tagged release publishes, as a downloadable artefact:

- RISCOF report (pass/fail per test, per config)
- Co-simulation instruction counts
- `riscv-formal` results
- Functional coverage report
- Synthesis: LUT/FF/BRAM/DSP, achieved `f_max`, critical path
- Benchmark table with history
- `platform.lock`

“The open academic core that ships with its own compliance evidence” is only a claim if the evidence is downloadable. This bundle is the claim.

29. FPGA implementation

29.1 Targets

Board	Role	Phase
Verilator	primary development target; counts as a board	2
KC705 (Kintex-7, DDR3, JTAG)	primary hardware target	2
Genesys 2	secondary, if available	4
ZCU104 / Zynq	ARM PS gives an easy host/DMA path for ML	5+

A board port is exactly four files (NFR-8): `board.yaml`, `board.xdc`, `board_top.sv`, `openocd.cfg`.

29.2 Bring-up order

Do these in order. Skipping ahead is how a week disappears.

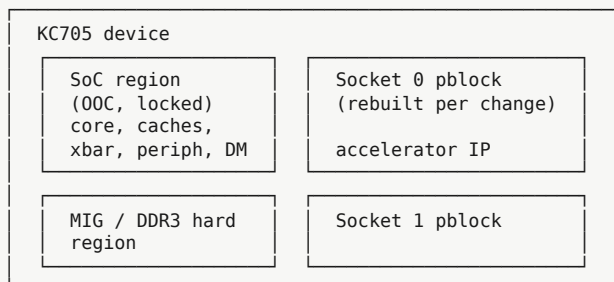
- | | |
|---|----------------------------------|
| 1. ☐ Bitstream builds, LED blinks from a counter | → clocks and PLL are alive |
| 2. ☐ UART transmits a fixed character | → IO and pin constraints correct |
| 3. ☐ UART echo | → RX path, clocking correct |
| 4. ☐ JTAG IDCODE reads back | → DTM alive |
| 5. ☐ OpenOCD connects, DM reports halted | → debug module alive |
| 6. ☐ Read/write on-chip SRAM over JTAG | → system bus access alive |
| 7. ☐ Core executes from boot ROM, GPIO toggles | → the core runs |
| 8. ☐ MIG calibration completes, DDR read/write over JTAG | → DRAM alive |
| 9. ☐ "Hello World" over UART from C | → full stack alive |
| 10. ☐ gdb load + breakpoint + step on an arbitrary ELF | → FR-9 met |
| 11. ☐ CoreMark runs and reports | → NFR-2 measurable |
| 12. ☐ Accelerator socket loopback test | → integration path alive |

29.3 Turnaround strategy

A full KC705 build is tens of minutes. That is the lab's real day-to-day bottleneck, and three things mitigate it:

1. **Verilator first.** Most functional bugs are found in simulation. The FPGA is for performance and for proving it is real.
2. **Out-of-context synthesis of the frozen SoC.** Because the socket interface is frozen (\$20), the infrastructure can be synthesised once and reused; only the accelerator partition rebuilds.
3. **Incremental implementation** with a locked placement checkpoint for the SoC region.

29.4 Floorplan sketch



30. Area, timing and power budgets

Design targets for Phase-0 planning. Replaced with measurements at the end of Phase 2; **keep the targets alongside the measurements** so the estimates get calibrated for the next project.

Block	LUT target	FF target	BRAM	DSP
S1-Core (pipeline, regfile, CSR)	12 k	6 k	0	4
I\$ 16 KB + D\$ 16 KB	4 k	2 k	16	0
MMU/PTW (Phase 5)	3 k	2 k	2	0
AXI crossbar 256-bit	12 k	8 k	0	0
Peripherals + AXI-Lite	5 k	3 k	2	0
Debug module + DTM	3 k	2 k	1	0
Platform total (S1-AI, no accel)	~45 k	~25 k	~24	4
KC705 (XC7K325T) available	203 k	407 k	445	840
Headroom for accelerators	~75%			

Timing target: ≥ 50 MHz (NFR-1), stretch 100 MHz. Expected critical paths, in likely order:

1. D\$ tag $\rightarrow$ hit $\rightarrow$ forward $\rightarrow$ ALU
2. CSR read $\rightarrow$ decode $\rightarrow$ operand mux
3. AXI crossbar arbitration at 256-bit
4. Completion buffer head comparison $\rightarrow$ retire

Track achieved $f_{\max}$ per nightly build and plot it. A gradual decline over a semester is the normal failure mode and is only visible if it is plotted.

Part VII — MEDS-S1 as a research platform

Parts I-VI describe a SoC. This part describes why it will produce publishable work.

31. Measurement infrastructure

Three levels, all shipped with the platform, all required for a defensible result:

Level	Mechanism	Answers
Application	<code>libs1_perf</code> , <code>PERF_BEGIN/END</code>	end-to-end latency, throughput, energy proxy
System	<code>mhpmcounter</code> events (§12)	where the cycles went; memory vs compute vs offload
Accelerator	<code>socket</code> <code>PERF_CYCLES</code> / <code>PERF_STALLS</code> (§20.2)	accelerator utilisation and its own stalls

An “energy proxy” in v1.0 means cycles × a per-block activity estimate — not a measurement. **Say so in every report.** Real energy measurement needs board instrumentation and is a separate project; claiming measured energy without it is the fastest way to lose a reviewer.

Measurement infrastructure — why a speedup number is not enough

APPLICATION	<code>libs1_perf</code> · <code>PERF_BEGIN/END</code>	end-to-end latency, throughput
SYSTEM	<code>mhpmcounter</code> events	where the cycles went: memory vs compute vs offload
ACCELERATOR	<code>socket</code> <code>PERF_CYCLES</code> / <code>PERF_STALLS</code>	accelerator utilisation and its own stalls

Attribution — the table that turns a number into a finding

<code>mxif_busy_cycles</code> low, total high	→ the accelerator is idle — software or offload path
<code>mxif_issue_stall_cycles</code> high	→ the core cannot feed it — consider loose coupling
<code>axi_read_latency</code> / beats high	→ memory-bound — look at tiling and double buffering
<code>axi_arb_stall_cycles</code> high	→ bus contention — check the bandwidth budget
<code>busy</code> high, <code>speedup</code> still low	→ genuinely compute-bound — the honest bottleneck

Only the last row is about the accelerator’s design. A thesis that cannot rule out the other four has not measured anything.

Figure 18 — Measurement infrastructure and the attribution table.

32. Reference workload suite

Frozen in Phase 2, versioned, never silently changed. Two tiers.

32.1 Tier A — architectural benchmarks (comparability with the outside world)

Benchmark	Reports	Why
CoreMark	CoreMark/MHz	the standard embedded comparison point
Embench-IoT	per-benchmark cycles	modern, less gameable than Dhrystone
Dhrystone	DMIPS/MHz	only for comparability with published cores

32.2 Tier B — edge AI and healthcare kernels (the lab's domain)

Each ships as scalar C reference + test vectors + expected output, so any accelerator can be scored against the same baseline.

Workload	Domain	Shape	Stresses
ECG arrhythmia classification	healthcare	1-D CNN, ~50 k params	1-D conv, streaming input
EEG seizure detection	healthcare	small CNN + LSTM	recurrent state, mixed layers
PPG heart-rate estimation	healthcare	FFT + peak detect	DSP, not NN — good contrast
Medical image segmentation tile	healthcare	U-Net tile, INT8	2-D conv, transposed conv, memory-bound
Keyword spotting	edge AI	DS-CNN, ~40 k params	depthwise conv — a very different dataflow
Anomaly detection	edge AI	autoencoder, MLP	dense GEMM
Person detection	edge AI	MobileNet-tiny INT8	depthwise + pointwise, the standard edge case
MEDS-V kernel set	vector	memcpy, saxpy, FIR, GEMM	already exists in RVV/examples/04-workloads

Why this set: it spans **1-D and 2-D convolution, depthwise and dense, DSP and NN, compute-bound and memory-bound**. An accelerator that helps everything here is genuinely general. One that helps only dense GEMM will be visibly narrow — which is a legitimate result, stated honestly.

32.3 The rule

Every workload has a scalar C reference that runs on S1-Base and produces bit-exact expected output. Without it there is no baseline, and without a baseline a speedup number means nothing.

33. Thesis project templates

33.1 What a student receives on day one

- A working SoC in Verilator and on KC705
- The socket or MXIF interface, frozen and documented

- A conformance testbench their design must pass
- A driver template with cache maintenance already correct
- The workload suite with scalar baselines already measured
- `libs1_perf` and the counter set
- A report template with the required tables of §34

33.2 What they deliver

Deliverable	Form
Accelerator RTL	passes conformance TB, lint clean, in <code>accelerators/<name>/</code>
<code>accel.yaml</code>	ID, version, register map, bandwidth demand (§22.4)
Driver + example application	in the repo, in CI
Unit + integration testbenches	in CI
Measured results	§34 tables, reproducible by <code>make</code>
Documentation	<code>README.md</code> + the thesis chapter

33.3 Candidate projects, sized

Project	Coupling	Effort	Novel contribution
INT8 systolic conv engine	socket	MS, 1 yr	dataflow choice, tiling study
Sparsity-aware conv engine	socket	MS/PhD	pruning-aware hardware, honest overhead accounting
Depthwise-separable engine	socket	MS	the dataflow mismatch nobody reports
Winograd / FFT convolution	socket	MS/PhD	numerics vs area trade-off
Activation + quantisation unit	MXIF	MS, 1 sem	tight-coupling case study; good contrast project
Posit arithmetic unit	MXIF	MS/PhD	alternative numerics, end-to-end accuracy
ECG-specific fused pipeline	socket	MS	domain specialisation vs generality
Approximate-compute MAC array	socket	PhD	accuracy/energy Pareto on real workloads
RVV extension of MEDS-V	MXIF	PhD	ratified-ISA vector work with compliance evidence
Branch predictor / cache study	core	MS	uses the frozen frontend interface (§6)
Energy-efficiency instrumentation	platform	MS	turns the §31 proxy into real measurement
NPU compiler / kernel library	software	MS/PhD	the software half nobody does

Note the last three. Not every thesis needs to be a datapath. Platform, measurement and compiler work are undersupplied in most labs and are often more citable.

33.4 The proposal checklist

No accelerator project starts without these four answered in writing:

1. **Which coupling mechanism, and why** (§21 flowchart).
2. **The §22.4 bandwidth calculation.** Required bandwidth vs available. If required > available, the proposal is about data movement, not arithmetic — say so.
3. **Which workloads from §32 it targets**, and the measured scalar baseline for each.
4. **What the honest negative result would look like.** A student who cannot describe how their idea might fail has not understood it yet.

34. Comparison methodology

Every result from this platform reports the same table. This is what makes a 2027 thesis comparable to a 2031 thesis, and it is the single highest-value thing a supervisor can standardise.

34.1 Required reporting table

Field	Example	Notes
Platform version	meds-s1 v1.2.0	from platform.lock
Config	s1_ai	§5.3
Board, clock	KC705, 100 MHz	
Workload, version	ecg_cnn v1.0	frozen suite (§32)
Baseline cycles	12,304,551	scalar C on the same config
Accelerated cycles	912,431	
Speedup	13.5×	end-to-end, not kernel-only
Accelerator utilisation	78.0%	socket PERF_CYCLES
Accelerator stall fraction	14.1%	socket PERF_STALLS
DRAM traffic	1.41 MB	axi_read_beats + axi_write_beats
Bus contention	2.4%	axi_arb_stall_cycles
Area	LUT / FF / BRAM / DSP	accelerator only, and platform total
f_max	152 MHz	accelerator OOC
Energy proxy	cycles × activity estimate	label as a proxy, never as measured

34.2 Rules

1. **End-to-end, always.** Kernel-only speedup excludes DMA, cache maintenance and invocation overhead — precisely the costs the platform exists to expose. Report kernel-only additionally if you like, never instead.
2. **Same config for baseline and accelerated.** Comparing an accelerated S1-AI against a baseline on S1-Nano is not a comparison.
3. **Report the negative cases.** A workload where the accelerator does not help is a finding. Suppressing it is the thing reviewers catch.
4. **Reproducible by make.** Every number in a thesis regenerable by one command, at a tagged commit, with platform.lock. This is also how a supervisor checks a result without rerunning the

student's whole environment.

5. **Area excludes the platform.** Report accelerator area separately from platform area, then both.

Appendices

Appendix A — Configuration matrix

See §5.3. Every config elaborates in CI on every merge. **A config not in CI does not exist.**

Appendix B — Default memory map

Region	Base	Size	Cacheable	Idempotent	Order	Atomic	Align
Debug ROM	0x0000_0000	4 K	no	yes	rvwmo	none	natural
Boot ROM	0x0000_1000	32 K	I-only	yes	rvwmo	none	natural
CLINT	0x0200_0000	64 K	no	no	strong	none	natural
PLIC	0x0C00_0000	4 M	no	no	strong	none	natural
UART0	0x1000_0000	4 K	no	no	strong	none	natural
SPI0	0x1000_1000	4 K	no	no	strong	none	natural
GPIO0	0x1000_2000	4 K	no	no	strong	none	natural
Timer0	0x1000_3000	4 K	no	no	strong	none	natural
Socket 0 MMIO	0x2000_0000	64 K	no	no	strong	none	natural
Socket 1 MMIO	0x2001_0000	64 K	no	no	strong	none	natural
On-chip SRAM	0x4000_0000	256 K	yes	yes	rvwmo	lrsc, amo	any
DRAM (cached)	0x8000_0000	1 G	yes	yes	rvwmo	lrsc, amo	any
DRAM (uncached alias)	0x1_0000_0000	1 G	no	yes	strong	amo	natural

Generated from `soc.yaml`. This table is documentation of the default; the file is the truth.

Appendix C — CSR list

Machine: `mvendorid`, `marchid`, `mimpid`, `mhartid`, `mstatus`, `misa`, `medeleg`, `mideleg`, `mie`, `mtvec`, `mcounteren`, `mscratch`, `mepc`, `mcause`, `mtval`, `mip`, `menvcfg`, `mcountinhibit`, `mcycle`, `minstret`, `mhpmcounter3–15`, `mhpmevent3–15`, `pmpcfg0–3`, `pmpaddr0–15`.

Supervisor (present, stubbed in v1.0): `sstatus`, `sie`, `stvec`, `scounteren`, `sscratch`, `sepc`, `scause`, `stval`, `sip`, `satp`.

User (read-only shadows): `cycle`, `time`, `instret`, `hpmcounter3–15`.

Debug: `dcsr`, `dpc`, `dscratch0`, `dscratch1`, `tselect`, `tdata1`, `tdata2`, `tdata3`.

MEDS-S1 platform: `MEDS_S1_PLATFORM_VERSION` (read-only, custom range, allocated in `extensions/REGISTRY.md`).

External (coprocessor-owned, routed via the §10.2 external CSR port): MEDS-V's `vstart`, `vxsat`, `vxrm`, `vcsr`, `vl`, `vtype`, `vlenb`.

Appendix D — RTL naming conventions

Rule	Example
Module prefix	<code>meds_sl_*</code> , <code>sl_core_*</code> , <code>meds_v_*</code>
Clock, reset	<code>clk_i</code> , <code>rst_ni</code> (async assert, sync de-assert, active low)
Direction suffix	<code>_i</code> , <code>_o</code> , <code>_io</code>
Handshake	<code>_valid</code> , <code>_ready</code> ; valid must not depend combinationally on ready
Active low	<code>_n</code> suffix
Parameters	UPPER_SNAKE
Types	<code>lower_snake_t</code>
Enums	UPPER_SNAKE_E values, <code>lower_snake_e</code> type
Generated files	header comment <code>DO NOT EDIT – generated from soc.yaml</code>
Every file	SPDX header, Apache-2.0, copyright Maktab-e-Digital Systems Lahore

Enforced by Verible in CI, not by review comments.

Appendix E — Glossary

Term	Meaning
AXI4 / AXI4-Lite	ARM bus protocols; full and simplified
BTFN	Backward-Taken Forward-Not-taken static branch prediction
CB	Completion buffer (§9)
CDC	Clock domain crossing
CLINT	Core-Local Interruptor: software and timer interrupts
CV-X-IF	OpenHW eXtension Interface; MXIF is a profile of it
Loose coupling	Accelerator attached as a memory-mapped device with DMA (§20)
MXIF	MEDS eXtension InterFace (§19, INTERFACES.md §1)
OOC	Out-of-context synthesis
PLIC	Platform-Level Interrupt Controller: external interrupts
PMA	Physical Memory Attributes (§11)
PMP	Physical Memory Protection (§11)
PTW	Page Table Walker (§16)
RISCOF	RISC-V Compatibility Framework — runs the architectural tests
RVFI	RISC-V Formal Interface — the trace port (§28.2)
Sail	The formal RISC-V ISA model; golden reference for RISCOF
Spike	The reference RISC-V ISA simulator; golden reference for co-simulation
Tight coupling	Accelerator invoked by an instruction over MXIF (§19)
Zicbom	Cache Block Management Operations extension

Revision history

Version	Date	Change
0.1	2026-08-02	Initial <code>design_doc.md</code>
0.2	2026-08-03	This document. Name ratified as MEDS-S1 / S1-Core. Adds two-phase MXIF completion (§9.2, §19.2, INTERFACES.md §1.4a) — without it, offload-at-retire would have blocked in-order retire for the duration of every coprocessor operation and eliminated the decoupling MEDS-V depends on. Adds Part VII (research platform), §22 (ML data movement), §32 (workload suite), §34 (comparison methodology).

This document is DRAFT until the Phase-0 design review ratifies it. INTERFACES.md §1, §2, §5 and §9 are the normative parts; everything here is explanatory except where marked normative.